

THE KANSAS UNIVERSITY QUARTERLY

COMMITTEE OF PUBLICATION

E. H. S. BAILEY

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E. MILLER

S. W. WILLISTON

V. L. KELLOGG, *Managing Editor*

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KANSAS MOSASAURS, PART I.	S. W. Williston and E. C. Case		
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JOURNAL PUBLISHING HOUSE,
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KANSAS PTERODACTYLS.

BY S. W. WILLISTON.

PART I, WITH [PLATE I](#).

The first American species of the singular group of extinct Mesozoic reptiles variously known as Ornithosaurs, Pterosaurs or Pterodactyls was described by Marsh from a fragmentary specimen obtained in 1870, by the Yale College Expedition in Wallace County, Kansas. About a dozen other specimens were obtained by a similar expedition the following year in charge of Professor Marsh, or by Professor Cope, and were described by these authors shortly afterward. By far the largest number of known specimens, however, other than those in the Kansas University Museum, were obtained during the years 1874, '75, '76 and '77 by parties of which Professor Mudge, Dr. H. A. Brous, E. W. Guild, George Cooper and myself were the members, and it was from these specimens that most of the published characters were derived. Many of these specimens are necessarily fragmentary ones, still the material now in the Yale College Museum is ample to elucidate everything of interest concerning these animals.

During the past few years, the Museum of Kansas University has been enriched by a series of excellent specimens of these animals, obtained from the same regions, specimens that permit the solution of most of the doubtful characters and throw not a little light on the affinities of the Kansas forms.

The species hitherto named are as follows:

PTERANODON.

Pteranodon Marsh, Amer. Journ. Sci. xi, p. 508, June 1876; and xii, p. 479, Dec. 1876; xxiii, p. 253, April,

1882; xxvii, p. 423, May, 1881; Williston, Amer. Naturalist, xxv, p. 1174, Dec. 1891

Pteranodon occidentalis.

Pterodactylus Oweni Marsh, Amer. Journ. Sci. i, p. 472, June 1871, Sep. p. 16 (nom. preoc).

Pterodactylus occidentalis Marsh, Amer. Journ. Sci. iii, p. 242, April 1872, Sep. p. 1; Cope, Cretac. Vert. p. 68, pl. vii, ff. 5, 6.

Ornithocheirus harpyia Cope, Proc. Amer. Phil. Soc. 1872, p. 471 (Cope).

This species was originally based upon the distal end of two wing-metacarpals, and teeth. In the following year, a fuller description was given of additional remains referred to the same species and renamed *P. occidentalis*.

Pteranodon ingens.

Pterodactylus ingens Marsh, Amer. Journ. Sci. iii, p. 246, April 1872, Sep. p. 6.

Pteranodon ingens Marsh, Amer. Journ. Sci. xi, p. 508, June 1876.

This species is based upon various bones of the wing-finger of several individuals, and three teeth.

Pteranodon umbrosus.

Ornithocheirus umbrosus Cope, Proc. Amer. Phil. Soc. 1872, p. 471.

Pterodactylus umbrosus Cope, Cret. Vert. p. 65, pl. vii, ff. 1-4.

Marsh (Amer. Journ. Sci. xii, p. 480, Dec. 1876) says this name is a synonym of *P. ingens*, published two days earlier. As this synonymy is not certain, and as Cope's species has been figured, I am not ready to accept his views.

Pteranodon velox.

Pterodactylus velox Marsh, Amer. Journ. Sci. iii, p. 247,
April 1872, Sep. p. 8.

Based upon the distal end of the right metacarpal of the wing-finger, and the proximal extremity of the adjoining first phalanx, two uncharacteristic parts of the skeleton, Marsh to the contrary notwithstanding. It is doubtful whether the direct comparison of the types will suffice to determine the species with certainty. "Both of the bones are somewhat distorted by pressure."

Pteranodon longiceps.

Pteranodon longiceps Marsh, Amer. Journ. Sci. xi, p.
508, June 1875; xxvii, p. 424, pl. xv, May 1884.

Based upon a somewhat defective skull, without other bones. There is no evidence whatever that the species is distinct from the preceding.

Pteranodon comptus.

Pteranodon comptus Marsh, Amer. Journ. Sci. xi, p. 509,
June 1876.

Based upon wing-bones of three individuals. The description is meagre.

Pteranodon nanus.

Pteranodon nanus Marsh, Amer. Journ. Sci. xxi, p. 343,
April 1881.

Based upon various remains of one individual; the humerus, alone, is recognizably described.

NYCTODACTYLUS.

Nyctosaurus Marsh, Amer. Journ. Sci. xii, p. 480, Dec.
1876. (nomen preoc.^[1]).

Nyctodactylus Marsh, Amer. Journ. Sci. xxi, p. 343,
April 1881: *ibid.* xxvii, p. 423, May 1884.

Nyctodactylus gracilis.

Pteranodon gracilis Marsh, Amer. Journ. Sci. xi, p. 508,
June 1876.

Nyctosaurus gracilis Marsh, Amer. Journ. Sci. xii, p.
480, Dec. 1876.

Nyctodactylus gracilis Marsh, Amer. Jour. Sci. xxi, p.
343, April 1881.

PTERANODON.

Skull.

Fragmentary portions of the skull of *Pteranodon* are not at all rare in the Kansas chalk; but it is exceedingly seldom that a complete, or even approximately complete specimen is found. Their great length and slenderness, together with the extensive pneumaticity of the bones, render their preservation, as a whole, a thing of great rarity. Probably the most nearly perfect one yet known is now in the Museum of Kansas University. It was discovered the past summer by Mr. E. C. Case, a member of the University Geological Expedition. The specimen was carefully cleaned on its upper surface, as it lay in the chalk, and then imbedded in plaster before removal. The surface now exposed was the under one, which surface is, almost invariably, better preserved and less distorted than the upper one in these animals. A figure of this specimen is given in Plate I. The only portion restored is that indicated by the line in the lower jaw; it is possible that this part of the symphysis may not be exactly as it is drawn. Other, incomplete, specimens in the Museum confirm the outlines, except in the occipital crest, which is not present. As stated by me in the *American Naturalist* (*l. c.*), the type specimen of *Pteranodon*, also collected by myself, was incomplete, and the figures of it, as given by Marsh, are faulty.

The elements of the skull are all so firmly united that they can not be distinguished. There are no indications whatever of a horny sheath enclosing the jaw, and it is improbable that the covering of these parts was essentially different from that in the slender jawed *Pterodactylidae*. In texture, the maxillaries are fine-grained, and wholly without the vascular

foramina found in the corresponding bones of birds. The bones are composed of two thin and firm plates, separated by cavities which are bounded by irregular walls of bony tissue. In the compression from which all the Pterodactyl bones have suffered more or less, the greater resistance of these walls has caused irregularities upon both the outer and the inner surfaces. At the borders of the bones, where the thickness has been greater, the roughening is not observed.

Seen from above, the skull is narrow, as stated by Marsh; but, contrary to his statement, there is not a sharp ridge extending along the upper border. This border is obtuse and rounded, and in the frontal region, flattened. The sagittal crest is large, but not nearly so large as it is figured by Marsh, the restored outline of whose figure is undoubtedly wrong. The texture of the bone forming the crest is materially different from that of the remaining bones of the skull. The bone is more roughened, and less firm. There is a well-developed ring of sclerotic ossifications. In the specimen figured, the separate plates measure from six to eight millimeters in diameter. They were not imbricated, as in the *Pythonomorpha*, but have a similar dense texture. There is a superior temporal arch, bridging over a small opening leading downward to the inferior temporal fossa. The following measurements will give the principal dimensions of this specimen.

Length from tip of premaxillary to occipital condyle	680 millim.
Extreme length of skull	780
Extent of crest beyond orbit	145
Greatest diameter of orbit	65
Antero-posterior diameter of nasal opening	135
Length of quadrate	120
Width of lower jaw at articulation	22

Pubis.

In a previous paper on the anatomy of *Pteranodon*,^[2] I stated that I had never seen the so-called “prepubic bones.” Since that time, however, an excellent specimen of them has been discovered among our material. The

specimen of which they are a part consists of the larger portion of the skeleton, and is perhaps conspecific with the one to which the described pelvis belongs. The figure given herewith will convey a good idea of their shape. The bones of the two sides are firmly co-ossified, and have been pressed nearly flat; the figure represents them as they are spread out in one plane. The bone is very thin throughout, with a slight thickening at the ischial (*a*) attachment only. Lying contiguous with the anterior projection, is a slender ventral rib (*b*). It is possible that the curvature of this bone may be inward, rather than outward.

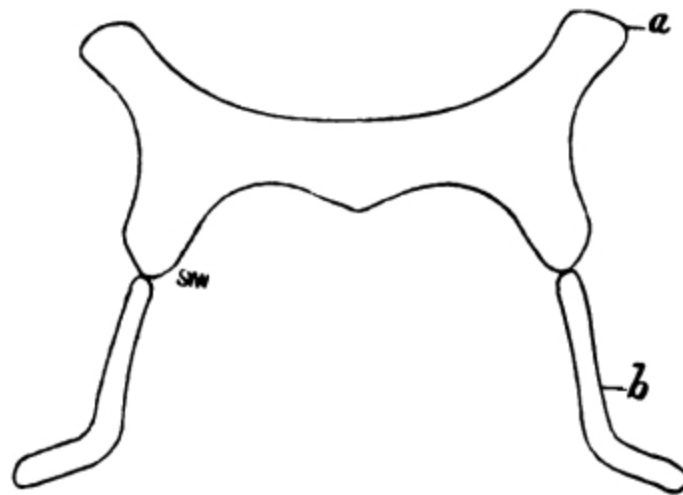


FIG. 1.

This peculiar structure of the pubis (I believe it represents the pubis, and not the prepubis), seems to be quite similar to that which obtains in the genus *Rhamphorhynchus*, and, perhaps also, in *Pterodactylus suevicus* (*Cycnorhamphus* Seeley), and very different from that found in other species of *Pterodactylus*.

The principal measurements of the above described specimen are as follows:

Antero-posterior expansion	40 millim.
Length of symphysis	14
Expanse of the united bones, as flattened	90

NYCTODACTYLUS.

The type species of this genus was described as follows by its author (loc. cit. supra):

“One of the smallest American species yet found is represented in the Yale Museum by several bones of the wing, a number of vertebrae and the nearly complete pelvis. The wing-bones preserved are elongated and very slender. The pelvis is unusually small, and there are five vertebrae in the sacrum. The last of the series indicates that the tail was short. The following are the principal measurements of this specimen:

Length of ulna	187 millim.
Length of metacarpal of wing-finger	300
Antero-posterior diameter of outer condyle at distal end	15
Transverse diameter of shaft, above condyles	13
Length of first phalanx of wing-finger	347
Extent of five vertebrae of sacrum	57

This species, which may be called *Pteranodon gracilis*, was about two-thirds the size of *P. velox* Marsh. It probably measured about ten feet between the tips of the expanded wings.”

In the December number of the same volume of the American Journal of Science, he described the genus as follows:

“A second genus of American Pterodactyls is represented in the Yale Museum by several well preserved specimens. This genus is nearly related to *Pteranodon*, but may be readily distinguished from it by the scapular arch, in which the

coracoid is not co-ossified with the scapula. The latter bone, moreover, has no articulation at its distal end, which is comparatively thin and expanded. The type of this species is *Pteranodon gracilis* Marsh, which may now be called *Nyctosaurus gracilis*. It was a Pterodactyl of medium size, measuring about eight to ten feet between the tips of the expanded wings."

The specific description of this species rests solely upon the measurements; the other characters given are not only vague, but are also common to all the known species. The generic description, as it is seen, is based upon the structure of the coraco-scapula. It will also be observed that the characters are not drawn from the type specimen, as that did not include this part of the skeleton, according to the author's statement. Of these two characters, the non-ossification of the coracoid and scapula is a somewhat doubtful one, as the same character may or may not occur in allied species, as, for example, in the species of *Rhamphorhynchus* (*R. Muensteri* Goldf.) described by the author himself. So incomplete and unsatisfactory are the characters thus given that Zittel, in his Handbuch, dismisses the genus with the brief remark, "noch unbeschrieben."

Nevertheless, from the peculiar form of the scapula, and from my recollection of the specimens upon which the genus was based, I believe I have determined with certainty an excellent specimen in the Snow Museum of Kansas University as a member of it, and here give a sufficiently complete description to place the genus on a more secure foundation.

This specimen was collected by Professor E. E. Slosson, of Wyoming University, while a member of my party in western Kansas the past season. It was partly exposed upon a gently sloping surface of firm yellow chalk on the Smoky Hill river, in the vicinity of Monument Rocks. Originally, the nearly complete skeleton must have been preserved, but a number of the bones had been either wholly or partially washed away, in some cases leaving their imprint in the chalk. The bones uncovered, and now lying upon the chalk slab nearly in their natural relations, are a humerus, both radii and ulnae, a pteroid, the two carpals of one wrist, both wing metacarpals, a first and a last wing phalanx, both coraco-scapulae, the posterior part of the lower jaws, ilium, femur, sternum, numerous ribs and vertebrae. The two coraco-scapulae lie with their scapular ends nearly

touching, and their coracoid ends separated by a space equivalent to the width of the sternal articulation. The two elements appear to have been imperfectly united and were probably not co-ossified. The inferior border of the coracoid, near the humeral articulation, has a greater expansion than is found in *Pteranodon*; its shaft is more rounded and less rugose, lacking especially the strong muscular markings upon the external surface. The articular surface does not appear to differ materially from that in *Pteranodon*. The scapula is of nearly the same length as the coracoid, but is much less stout. It is a thin, spatulate bone, slightly expanded at the distal extremity, where the margin is rounded, and without the characteristic oblique articular facet. It has no supra-glenoid expansion or process on the posterior proximal border, but has its margin nearly straight or gently concave from the articulation to its extremity. The space included between the bones of the two sides as they lie is a nearly regular, oval one, measuring ninety-five millimeters in its greater, forty-five in its lesser diameter.

The sternum lies at a little distance from the coraco-scapulae. It is an extremely thin bone, with a stout anterior, styliiform projection, at the base of which, on either side, looking upward and outward, is the articular, trochlea-like surface for the sternal end of the coracoid. The width between these articular surfaces measures fifteen millimeters; the length of the process in front of the articulations is twenty-five millimeters. Immediately posterior to the articular surfaces, the bone expands nearly at right angles to the longitudinal axis to a width of about sixty millimeters. The thin lateral margins are nearly parallel with the longitudinal axis, and show three shallow emarginations between the four costal articular projections. The hind angles are nearly rectangular. The bone, as preserved, is only shallowly concave, and shows no true keel, though a more pronounced median convexity towards the front doubtless subserved the function of a carina in part.

The left humerus lies in position, and is especially characterized by its enormous deltoid crest (radial crest of Marsh), though otherwise slender. This crest is further removed from the head of the bone than is the case in species of *Pteranodon*. It is directed somewhat downward, and has its distal, gently convex, border about twenty-five millimeters in extent, while the width of the process midway between the extremity and the base

measures but sixteen millimeters. The bicipital crest is also prominent. The bone is relatively shorter than in *Pteranodon*.

The humerus, as will be seen from the above description, and from the measurements given below, is remarkably like the same bone in *Pteranodon nanus*, as described by Marsh (*l. c. supra*), and but a little larger. In *P. nanus*, however, the coracoid and scapula are said to be firmly co-ossified, and the scapula has of course a different structure.

The skull has been, unfortunately, almost wholly washed away, a fragment of the cranial wall and the posterior part of the lower jaws alone remaining. It is impossible, hence, to say much concerning this part of the anatomy. The lower jaws show a different structure from that in *Pteranodon*. As they lie in their natural position, the width at the condyles is about twenty-four millimeters. The angular is less produced posterior to the articulation than in *Pteranodon*, indicating a less elongated and less powerful mandibular portion, an indication further borne out by the slenderness of the rami. The impression in the chalk shows the symphysis to begin ninety millimeters from the articulation. The width at this place could not have exceeded sixteen millimeters; and the entire length of the lower jaws could hardly have been more than one hundred and twenty-five millimeters. In the parts preserved, measuring seventy-five millimeters, there are no indications of teeth; yet it is not impossible that there may have been teeth in the anterior portion of the dentary, as in some species of *Pterodactylus*. I hardly think it probable, however.

There are seven cervical vertebrae preserved, apparently the full complement, as in *Pteranodon* and other members of the order. They differ in no especial respect from the corresponding vertebrae of *Pteranodon*, and, apparently, of *Pterodactylus*. The imperfectly anchylosed, possibly free, atlas shows three pieces, the odontoid process and the two slender lateral pieces. The lateral pieces are entirely free, with a thickened base and a slender, curved upper portion. The odontoid is gently concave in front, and seems to be imperfectly ossified with the axis; it occupies the lower part of the articulation, corresponding to the hypapophysis of the Pythonomorpha. The axis is the shortest of the remaining vertebrae, and has a well developed spine. The centrum is strongly convex behind, as are the remaining centra of the series. The following five vertebrae decrease gradually in length. The anterior ones have only a thin ridge or plate for the

neural spine; the seventh, however, has a neurapophysis of some length. They are all, as is usually the case, somewhat distorted from pressure. The under side is flattened, apparently gently concave longitudinally, and with a lateral ridge terminating in an obtuse hypapophysis at each inferior hind angle.

In his discussion of the Pterosauria, Zittel says concerning the vertebrae: "zwischen oberen Bogen und Centrum ist keine Suture zu bemerken." Handbuch, iii, p. 776. In this he is in error, so far as the American forms are concerned. It is usually the case in the Kansas specimens of both genera that the neural arch of the post-cervical vertebrae is wholly or in part detached from the centrum, showing a sutural, and not ankylosed union in life. The centra of twelve vertebrae are preserved, in the present specimen, from the region back of the neck; in only five of them are the neural arches in any way attached. Three of these are evidently anterior thoracic, judging from their structure and the position in which they lie. The shortest of them, to which was attached a very large rib, and which was lying in front of the scapulae, may represent the first thoracic vertebra (*a*). Its centrum is fully as wide as long, is flat on the under surface, and has a large, stout, horizontal parapophysis near the anterior end. Just above this process for the attachment of the head of the rib, and separated by a deep notch, is a much more elongated, horizontal diapophysis for the tuberculum. The cup of the centrum is shallowly concave; the transverse, shallowly U-shaped ball is only a little convex.

Two other vertebrae (*b*), found close by the one just described, and possibly one or the other contiguous with it, differ remarkably in having no, or a rudimentary, parapophysial process, and in having the diapophyses much shorter. It is not impossible that a slight expansion at the lateral margins of the ball may represent small parapophyses. In *Pteranodon* there are at least four vertebrae with dia- and parapophyses. In the other vertebrae from this region the diapophyses are yet shorter and the neural spine stouter and broader. The other centra preserved are all shaped somewhat like the half of a cylinder, and are a little longer than broad. They have no distinct cup or ball. In two of them there is a very long, recurved parapophysial process, as though formed by an ankylosed rib, on each side; they are probably lumbar vertebrae.

Most of the ribs are very slender; a few are moderately thickened; one only is very stout; its measurements are given below.

Length of lateral pieces of the atlas	7 millim.
Diameter of lateral pieces at the base	3½
Width of odontoid	4½
Height of odontoid	3
Length of axis	8
Height of axis	15
Length of third cervical vertebra	21
Length of fourth cervical vertebra	20
Length of fifth cervical vertebra	19
Length of sixth cervical vertebra	18
Length of seventh cervical vertebra	17
Height of seventh cervical (about)	15
Length of centrum, anterior thoracic vertebra (<i>a</i>)	6
Width of ball (<i>a</i>)	8
Expanse of parapophyses (<i>a</i>)	14
Expanse of diapophyses (<i>a</i>)	26
Width of neural canal (<i>a</i>)	3
Length of centrum, anterior thoracic vertebra (<i>b</i>)	8
Width of ball (<i>b</i>)	10
Expanse of diapophyses (<i>b</i>)	17
Height of neural spine (<i>b</i>)	20
Width of neural spine (<i>b</i>)	5
Length of rib (<i>c</i>)	45
Width of shaft (<i>c</i>)	5
Distance from center of capitulum to center of tubercle (<i>c</i>)	10
Length of coracoid	50
Antero-posterior diameter, sternal extremity	9
Length of scapula	45
Width of scapula at distal end	15
Length of humerus	80

Width through deltoid crest	24
Least diameter of shaft of humerus	13
Length of ulna	133
Width of ulna at distal extremity	22
Length of radius	130
Width of radius distally	15
Length of wing-finger metacarpal	220
Width of same metacarpal at proximal end	20
Diameter through condyles	15
Transverse diameter of shaft above condyles	10
Length of first phalanx, wing-finger	263
Width of same phalanx at proximal end	24
Width of same phalanx at distal end	15
Width of sternum	67
Length of rib borders	25
Length of femur	75
Diameter of head of femur	5
Diameter of femur through condyles	12
Length of pteroid bone	88

The principal dimensions of this species can be got at with considerable certainty. Although two of the wing-phalanges and the bones of the foot are wanting, yet the relative proportions of those present agree so closely with those of the corresponding bones in *Pteranodon*, that there can be but little possibility of error in assuming the same proportions for the missing ones. The position of the ilium and femur, as also the ribs, show that they hold their natural relations to the pectoral arch. The tail, alone, can not be got at.

Extreme expanse of wing-bones	2400 mm.	7 ft.	10 in.
Expanse of wings in life, approximated	2000	6	6
Length of head, estimated	150		6
Length of neck	128		5½
Length of trunk	165		6½

But one species has been described from the American Cretaceous smaller than the present one, *Pteranodon nanus* Marsh, in which the expanse of wings is given as not more than three or four feet. In this estimate the author is certainly in error. The size of the humerus, as given, is rather more than three-fourths that of the present species, and the expanse, hence, must be nearly five feet in life, or six feet as the bones lie outstretched.

As regards the specific determination of the present specimen, there must necessarily be some doubt until the species already named have been recognizably described. But three of the existing species can be taken into account, *N. gracilis*, *P. comptus* and *P. nanus*. That it can not be the last, has already been shown. In size, it agrees well with *P. comptus*, but the other characters throw no light upon the identity.

The measurements given of the type specimen of *N. gracilis* show the size to be materially greater,—a character, however, of subordinate value—greater slenderness, and a relatively shorter first wing-phalanx.

The relative lengths of wing-metacarpals, wing-phalanx and ulna in *N. gracilis* and the present specimen may be expressed as follows:

Length of wing-metacarpal	100	100
Length of first wing-phalanx	115.6	119.5
Length of ulna	62.3	60.4

It will be seen that not a single character has yet been given to distinguish the genus from *Pterodactylus*, and it is not at all impossible that it may prove to be the same; its location among the *Pteranodontidae* rests solely on the assumed absence of teeth, and that is a character yet wholly unknown.

The material now in the museum permits a fuller discussion of the relations and characters of this group of reptiles than has been hitherto

attempted. Originally, they were described as constituting a new order, a view still held by its author and no one else. Lydekker, in his *Paleontology and Catalogue* gives them a subordinal value; Zittel only a family value, though expressing doubt as to their subordinal rank.

It seems very probable that the genus *Nyctodactylus* has no teeth in the jaws; it agrees in *every other respect* with the genus *Pterodactylus*, so far as known. If the genus has teeth it must be united with *Pterodactylus*. Now, in not a few species of this genus, the teeth are confined to the anterior end of the jaws, and their entire absence, unaccompanied by other structural differences, will hardly constitute an order, or even family.

But, leaving aside *Nyctodactylus*, it is very much of a question whether the differences between *Pterodactylus* and *Pteranodon* are sufficient to locate them in different families, let alone different suborders.

The two genera have the following in common: Tail short. Skull with more or less elongated, pointed jaws, and very small upper and lower temporal fossae. Narial opening large, confluent with the pre-orbital foramen. Cervical vertebrae elongated, with rudimentary spinous processes. Fore and hind extremities, quite alike.

Pteranodon differs from *Pterodactylus*, so far as that genus is known, in the united coracoscapulae and pubes, both of which characters are found in *Rhamphorhynchus*.

The sole family characters remaining then, for *Pteranodon*, are, absence of teeth, a supra-occipital crest, and the articulation of the upper end of the scapula. Now it seems evident that to place the pteranodonts in a group equivalent to all the other pterosaurs is unwarranted, and any classification that will not show the more pronounced relationships with *Pterodactylus* is faulty. I would, therefore, propose the following:

Order Pterosauria.

Family Pterodactylidae,

subfamilies Pteranodontinae, Pterodactylinae.

Family Rhamphorhynchidae.

Family Ornithocheiridae.

As regards the geographical distribution of the Pteranodonts, they have hitherto been recognized only from Kansas, but I am firmly of the opinion that they occur in Europe, and, if so, it is very probable that the name *Pteranodon* must be eventually given up. In fact, a toothless form of Pterodactyl was described by Seeley as long ago as 1871, under the name of *Ornithostoma*. I cannot refer to his description at present, and can, therefore, give no opinion as to their identity. It seems certain that the peculiar form of the scapulae and their vertebral articulation^[3] occur among some of the European forms, which would strengthen the belief that *Pteranodon* is also an European genus.

In view of the above, the practice of the American text-books in Geology in introducing generic names of characteristic fossils as names of the geological horizons whence they come, is very reprehensible, in my opinion. Even the late edition of Leconte's Elements contains a long list of such names, the greater portion of which have been relegated to the limbo of synonymy by paleontologists. It is greatly to be desired that the name "Pteranodon Beds" shall not become established, so long as there is the least doubt of the validity of the name itself.

KANSAS MOSASAURS.

BY S. W. WILLISTON AND E. C. CASE.

PART I, CLIDASTES, WITH [PLATES II-VI](#).

The group of extinct Cretaceous reptiles known as the Mosasaurs or Pythonomorpha was defined by Cope, "to whom Science is so largely indebted for its present knowledge of this interesting order of reptiles" (Marsh), in 1869.^[4] Although some of the characters assigned by him to the order have since been shown to be inapplicable, and the group to have less value, yet his name, Pythonomorpha, has been generally retained. Lydekker and Zittel have assigned to the group a subordinal value, as has also Marsh, though under a different name. Owen rejected it entirely, and Baur, more recently,^[5] has united it with the Varanidae to form a super-family, as follows:

Suborder Platynota.

Super-family Varanoidea.

Families Mosasauridae, Varanidae.

Super-family Helodermatoidea.

Family Helodermatidae.

The group, whatever may be its rank or position, includes, so far, the following genera: *Mosasaurus* Conyb., *Liodon* Owen, *Platecarpus* Cope, *Clidastes* Cope, *Baptosaurus* Marsh, *Sironectes* Cope, *Plioplatecarpus* Dollo and *Hainosaurus* Dollo. *Pterycollasaurus* Dollo, founded upon *Mosasaurus maximilianus* Goldf., is omitted as doubtful. All of these genera, save *Plioplatecarpus* and *Hainosaurus*, have been recorded from North America, *Clidastes*, *Baptosaurus* and *Sironectes* being peculiar to this country. Of these latter three genera, however, *Clidastes* alone is well known; but this genus is suspected by Lydekker of being the same as the imperfectly known European *Geosaurus* Cuvier. Thus it seems that the

genera, or at least the most of them, have a wide distribution; *Platecarpus*, in fact, is said to occur in New Zealand.

In America, members of the group have been discovered in the Cretaceous deposits of New Jersey, Alabama, North Carolina, the upper Missouri region, Nebraska, Kansas and New Mexico. Probably nineteen-twentieths of all the known specimens, however, have been obtained in western Kansas. The material now in the University Museum, all from Kansas, comprises several hundred specimens of these animals, including, probably, the best ones known. It is upon this material that the following preliminary studies are chiefly based.

The genus *Clidastes*, as first described by Cope, was based upon two dorsal vertebrae of *C. iguanavus*, the type species, from New Jersey. Shortly afterward, however, he gave a full and careful generic description, as derived from an unusually good specimen of an allied species, *C. propython*, from Alabama. Only a little later, Marsh described a genus, which he called *Edestosaurus*, from Kansas, but without giving any real, distinctive differences from *Clidastes*, following the very reprehensible practice of naming supposed new forms in the hopes that future distinctive characters might be found. The genus *Edestosaurus* has been rejected by nearly all save the authors of the American text-books in Geology. It seems hardly necessary to point out the identity. The only distinctive character the author gave for his genus was the insertion of the pterygoid teeth, and even this character he modified later—"Palatine (sic) teeth more or less pleurodont."^[6]

This character, even were it real, is of very slight value; indeed it cannot be used to distinguish the species even.

Clidastes is, without doubt, one of the most highly specialized genera in the group, and, what is very interesting, is one of the latest. It occurs in Kansas in the uppermost part of the Niobrara beds, in the horizon so markedly characterized by the toothed birds. Both *Platecarpus* and *Liodon* occur, though in diminished numbers, almost to the very lowest portion, but *Clidastes* has never been found except towards the top. From measurements made the past season, the thickness of the beds in which these saurians occur cannot be less than six hundred feet.

The following species have been found in Kansas: none of them are known to occur elsewhere.

MOSASAURIDAE.

Mosasauridae Conybeare, in Cuvier, Ossem. Foss., 2nd ed., p. 338, 1824.

Clidastidae Cope, Extinct Batr. Rept. and Aves of N. Amer., Trans. Amer. Phil. Soc. xiv, p. 50, 1870.

Edestosauridae Marsh, Amer. Journ. Sci. xxi, p. 59, July 1878.

CLIDASTES.

? *Geosaurus* Cuvier, Ossem. Foss. 2nd ed., 328, 1824, (*fide* Lydekker.)

Clidastes Cope, Proc. Acad. Nat. Sci. Phil. 1868, p. 233; Ext. Batr. etc., p. 21, 1870.

Edestosaurus Marsh, Amer. Journ. Sci. i, p. 417, June, 1871.

C. cineriarum.

Clidastes cineriarum Cope, Proc. Amer. Phil. Soc., 1870, p. 583; Cret. Vert. etc. pp. 137, 266, pl. xxi, ff. 14-17; Bullet. U. S. Geol. Surv. Hayden, iii, p. 583.

C. dispar.

Edestosaurus dispar Marsh, op. cit. i, p. 447, June 1871; iii, pl. xi., June, 1872.

C. velox.

Edestosaurus velox Marsh, Amer. Journ. Sci. i. p. 450, June, 1871.

Edestosaurus pumilus Marsh, *ibid.* p. 452.

? *Clidastes affinis* Leidy, Proc. Acad. Nat. Sci., 1870, p. 4; Rep. U. S. Geol. Surv., Hayden, vol. i, p. 283, 1873.

? *Edestosaurus dispar* Marsh, op. cit. xix, pl. i, f. 1, Jan., 1880.

C. Wymani.

Clidastes Wymani Marsh, Amer. Journ. Sci. i, p. 451, June, 1871; iii, p. 202, April, 1872.

Edestosaurus Wymani Marsh, op. cit. iii, p. 464, June, 1872.

C. tortor.

Edestosaurus tortor Cope, Proc. Amer. Phil. Soc. Dec., 1871; Marsh, op. cit. iii, p. 464, June, 1872.

Clidastes tortor Cope, Cret. Vert. Rep. U. S. Geol. Surv., Hayden, vol. ii, pp. 48, 131, 265, pls. iv, f. i; xiv, f. i; xvi, ff. 2, 3; xvii, f. 1; xix, ff. 1-10; xxxvi, f. 3; xxxvii, f. 2; Bullet. U. S. Geol. Surv. Hayden, vol. iii, p. 583.

C. stenops.

Edestosaurus stenops Cope, Proc. Amer. Phil. Soc. p. 330, 1871; Marsh, Amer. Journ. Sci. iii, p. 464, June, 1872.

Clidastes stenops Cope, Cret. Vert. etc. pp. 133, 266, pls. xiv, ff. 4, 5; xvii, f. 7, 8; xviii, ff. 1-5; xxxvi, f. 4; xxxvii, f. 3; xxxviii, f. 3.

C. rex.

Edestosaurus rex Marsh, op. cit. iii, p. 462, pl. xxii, f. 1, June, 1872.

C. planifrons.

Clidastes planifrons Cope, Bullet. U. S. Geol. Surv. No. 2, p. 31, 1874; Cret. Vert. etc. pp. 135, 265, pls. xxii, xxiii.

C. Westii.

C. Westii Williston, n. sp. infra.

CLIDASTES VELOX.

A remarkably complete specimen, referred with considerable certainty to this species, was obtained by ourselves in western Kansas, (Butte Creek) in the summer of 1891. A brief preliminary description of the specimen was given by the senior author in Science, December 8, 1891. A more complete description is here given, which, it is believed, will be of service. The specimen is an unusually perfect one, being very nearly complete, and, as now mounted, shows the bones nearly all in the position in which they were found. The vertebral column is continuous, except in one place, where the tail had been bent up over the back; and complete, save at the very tip of the tail. The skull is complete, or very nearly complete, and has been restored nearly to the condition in life. Figures have been made of this portion of the skeleton, and will be given in a future communication. At present, it may be mentioned that the lacrymals are small, roughly irregular bones, and pointed at either extremity. There are no indications of transverse bones, as there are none in any other skull in the collection.

Cervical vertebrae.

ATLAS. The intercentrum is a small bone with three sides of nearly equal extent. The two upper, articular surfaces are gently concave, and meet in a rounded margin; the inferior surface is convex, both antero-posteriorly and transversely, with a roughened prominence in the middle. The lateral pieces have indistinctly separated facets for articulation with the odontoid, the intercentrum and the occipital condyle. The rather short, flattened lamina extends upward, backward and inward, approaching, but not reaching its fellow of the opposite side; it is somewhat dilated distally. Directed outwards and forwards, there is a stout styliiform process.

Axis. The neural spine of the axis is elongated antero-posteriorly. It is thin on the anterior portion, but stouter and longer at the posterior part. The large, stout odontoid process is united suturally, as is also the well-developed atlantar hypapophysis, which forms the anterior, inferior portion of the bone. The diapophyses are the smallest of the costiferous series, with only a small articular facet for the rib. The ball is strongly and evenly convex, with its greater diameter transversely. The hypapophysis is the largest of the series; it is suturally united with the stout, exogenous process of the centrum, and projects downward and backward; its distal extremity is roughened for ligamentous attachments.

The third cervical vertebra shows a well-developed zygosphenal articulation, and stout articular processes. The transverse process is small, only a little larger than that of the axis, though, unlike that, it is strengthened by a ridge continued from the anterior zygapophyses. The hypapophysis is smaller than that of the axis, but, like that, is directed downward and backward. The spine may be distinguished from that of any other vertebra by its stout, trihedral shape; it is directed rather more obliquely backward than in the following vertebrae.

The fourth cervical vertebra differs from the third in having stouter transverse processes; in the hypapophysis being directed more nearly downward, and in its smaller size; and in the spine being flattened antero-posteriorly toward the base.

The fifth cervical vertebra differs from the fourth in the broader spine, in the stouter transverse processes, and the smaller hypapophysis.

In the sixth cervical vertebra, the hypapophysis is reduced to a small ossification, scarcely longer than broad, directed downward. The spine has reached nearly the full width of those of the following vertebrae, though somewhat stouter above. The transverse processes are yet stouter.

In the seventh, or last, cervical vertebra the hypapophysis is wanting, or very rudimentary. The under part of the centrum shows a rounded ridge or carina, with a slight projection corresponding to the hypapophysis.

MEASUREMENTS OF THE CERVICAL VERTEBRAE.

1. Antero-posterior diameter of intercentrum of atlas	14 millim.
Transverse diameter of intercentrum	25
Antero-posterior diameter of lateral piece	20
Vertical extent of articular surface	17
Extent of lateral piece	35
Width of lamina above	16
2. Length of axis	43
Transverse diameter of ball	18
Vertical diameter of ball	17
Expanse of transverse processes	28
Elevation of spine above floor of neural canal	34
Antero-posterior extent of spine	50
3. Length of third cervical vertebra	37
Height of spine above floor of neural canal	36
Depth of hypapophysis below floor of neural canal	34
4. Length of fourth cervical vertebra	37
Height of spine above floor of neural canal	39
Depth of hypapophysis below floor of neural canal	35
5. Length of fifth cervical vertebra	37
Height of spine above floor of neural canal	42
Depth of hypapophysis below floor of neural canal	33
Transverse diameter of ball	17
Vertical diameter of ball	18
6. Length of sixth cervical vertebra	37
Height of spine above floor of neural canal	42
Depth of hypapophysis below floor of neural canal	30
Width of spinous process	26
7. Length of seventh cervical vertebra	37
Height of spine above floor of neural canal	46
Transverse diameter of ball	19
Vertical diameter of ball	20
Width of spinous process	27

Dorsal vertebrae.

There are thirty-five vertebrae between the cervicals and the first non-rib-bearing vertebra, to which the pelvis was, evidently, attached. The distinction between the cervicals and thoracics cannot be made from any characters they possess, as the seventh vertebra does not bear a distinct hypapophysis. Neither can it be said with certainty from this specimen which is the first thoracic vertebra, as the cervical ribs had, unfortunately, been displaced in the collection and preparation of the specimen. In another specimen, referred to *C. pumilus*, and which, as will be seen later, cannot be specifically distinguished from the present species, short cervical ribs were found attached to six vertebrae posterior to the atlas. That the eighth vertebra is a thoracic one is shown by the relation of the ribs in this specimen. Posteriorly there is no distinction, also, between the true thoracic vertebrae and those of the lumbar region. All the vertebrae anterior to the pelvis bear ribs, and will all be considered as dorsal vertebrae, the true thoracic vertebrae being restricted to those of which the ribs are elongated, and, probably, connected with the sternum.

In the anterior vertebrae of the series, the centra are subcarinate below, the obtuse, rounded ridge becoming less and less apparent until no indications of the keel can be seen, before the middle of the series. The transverse processes are stoutest, with a more elongated, sigmoid articular surface, with little or no constriction, and projecting only slightly beyond the stout articulating processes, in the anterior vertebrae. In the tenth or eleventh, the articular surface has become markedly smaller, more vertical, and less sigmoid in outline. Thence to the last, the articular surface for the ribs remains nearly the same. The process itself, however, becomes gradually more prominent and constricted, as the zygapophyses becomes smaller. The spinous processes increase slightly in length and breadth, and are only slightly oblique throughout. In length, the centra increase gradually. The vertical diameter of the ball increases gradually, while the transverse diameter remains more nearly the same.

MEASUREMENTS OF THE DORSAL VERTEBRAE.

1. Length of centrum to rim of ball	38 millim.
Transverse diameter of ball	20
Vertical diameter of ball	19

Height of spine above floor of neural canal	48
Extent of articular surface of transverse process	30
Width of spine	28
4. Length of centrum to rim of ball	41
Transverse diameter of ball	20
Vertical diameter of ball	20
Height of spine above floor of neural canal	48
11. Length of centrum to rim of ball	41
Vertical diameter of ball	22
Extent of articular surface of transverse process	16
Width of spine	32
15. Length of centrum to rim of ball	41
Transverse diameter of ball	21
Vertical diameter of ball	24
20. Length of centrum to rim of ball	42
Vertical diameter of ball	25
Height of spine above floor of neural canal	58
24. Length to rim of ball	41
Transverse diameter of ball	22
Vertical diameter of ball	23
Height of spine	49
28. Length to rim of ball	40
Vertical diameter of ball	24
Transverse diameter of ball	23
Height of spine	54
32. Length to rim of ball	38
Vertical diameter of ball	25
Transverse diameter of ball	24
35. Length to rim of ball	37

Caudal vertebrae.

Immediately following the thirty-fifth rib-bearing vertebra there is an abrupt change, the tubercular process for the rib giving place to an

elongated transverse process. From the position of the pelvis, it is evident that the ilia were attached to the first pair of these. Precisely this relation of pelvis to the vertebrae is found in such lizards as the Monitor and Iguana, and it is probable that such is the relation in all the Pythonomorpha. It will thus be seen that there are no distinctively lumbar vertebrae, if by such are meant free, non-costiferous, pre-sacral vertebrae. The vertebrae of these animals that have been so designated by writers are in reality basal caudal. A distinctive term for them—those with transverse, non-costiferous processes and without chevrons—is needed, and we propose, provisionally, the term *pygial*. There are seven in the present series, all characterized by elongated transverse processes, and not differing much from each other. The vertebrae lie in the matrix with the ventral aspect uppermost, concealing the spine and upper parts. The under surface is somewhat flattened, and, as in the preceding vertebrae, is gently concave antero-posteriorly. The transverse processes are elongate, stout towards the base, apparently all of nearly equal length, and directed gently backwards and downwards. In the anterior vertebrae the processes spring from near the front part: as the centra become shorter they arise from near the middle. In the last one of the series there are minute indications of chevrons.

MEASUREMENTS OF THE PYGIAL CAUDAL VERTEBRAE.

1. Length to rim of ball	36 millim.
Width of ball	25
Expanse of transverse processes	130
Width of transverse process near base	17
2. Length to rim of ball	33
3. Length to rim of ball	31
4. Length to rim of ball	29
5. Length to rim of ball	28
6. Length to rim of ball	27
Expanse of transverse processes	130
Width of ball	24
7. Length to rim of ball	27

The centra of those caudal vertebrae which have chevrons do not differ much in shape. They become less constricted, and, back of the middle of the series, are smoothly cylindrical in shape. The transverse processes decrease gradually in length, disappearing entirely in the twenty-fifth or twenty-sixth. The spinous processes are more or less incompletely preserved in the anterior vertebrae. They increase only gradually in length for the first twenty of the series, and are markedly oblique, with the posterior border stout, and the anterior border alate. With the twenty-sixth they begin to increase more rapidly in length, and have become more nearly vertical in position, and are thinner at each margin. In the thirty-fifth or thirty-sixth they attain their greatest length, and are here directed slightly forwards. Thence to the end of the tail, the length decreases gradually, and, in position, they are directed more and more obliquely backward. The chevrons are strongly oblique throughout the series and are firmly co-ossified with the centrum.

The tail, it is thus seen, has a broad, vertical, fin-like extremity, which, doubtless, aided much in the propulsion of the animal through the water.

There are sixty-seven vertebrae with chevrons present in the specimen, all continuous, except in one place. The last one is less than one-fourth of an inch in diameter, and shows that there had been yet another, possibly several more. Toward the base of the series the tail has been bent forwards over the back, and it is possible that, where the break occurs, there has been a vertebra lost. The measurements, however, do not seem to indicate any loss. The entire series of vertebrae was not less than sixty-eight, and probably not more than seventy, making for the entire vertebral series one hundred and seventeen to twenty.

MEASUREMENTS OF THE CHEVRON-BEARING CAUDAL VERTEBRAE.

1. Length to rim of ball	26 millim.
5. Length to rim of ball	24
Vertical diameter of ball	21
Transverse diameter of ball	24
10. Length to rim of ball	24

15. Length to rim of ball	24
Height of spine above floor of neural canal	40
Length of chevron	45
20. Length to rim of ball	23
Vertical diameter of ball	21
Transverse diameter of ball	22
25. Length to rim of ball	20
Height of spine	44
Width of spine at base	19
Width of spine at distal end	10
Length of chevron	85
Altitude of tail	112
30. Length to rim of ball	18
Vertical diameter of ball	17
Height of spine	57
Width of spine at base	19
Width of spine at distal end	9
Length of chevron	99
Altitude of tail	20
35. Length to rim of ball	16
Vertical diameter of ball	16
Height of spine	61
Length of chevron	97
Altitude of tail	122
40. Length to rim of ball	15
Vertical diameter of ball	15
Height of spine	54
Length of chevron	70
Altitude of tail	110
45. Length to rim of ball	14
Vertical diameter of ball	14
Height of spine	40
Length of spine	50
Length of chevron	58

Altitude of tail	93
50. Length to rim of ball	13
Length of spine	43
Length of chevron	55
Altitude of tail	73
55. Length to rim of ball	12
Length of spine	38
Length of chevron	42
Altitude of tail	63
60. Length to rim of ball	9
Length of spine	46
Length of chevron	25
Altitude of tail	50
66. Length to rim of ball	7
Length of chevron	10
Altitude of tail	20
67. Length	6

Ribs.

As has already been stated, the cervical ribs were displaced in the present specimen, and measurements of them cannot be given. In a smaller specimen, specifically indistinguishable from the present one, the entire cervical series is preserved with the ribs attached. The first, that articulating with the axis, is very short. The following ones are stouter, but increase only moderately in length, that of the sixth measuring only thirty-five millimeters, while that of the seventh is but a little longer. In the specimen of *C. velox* described, there is a detached cervical rib sixty-five millimeters in length; it probably belongs with the seventh.

The thoracic ribs are simple, somewhat flattened rods, moderately expanded at the proximal end. The greatest convexity is shown about the middle of the series, where the versedsine of the curvature is forty millimeters, the chord being one hundred and sixty. Posteriorly, the short ribs are only gently curved.

Lying by the side of the vertebral column, and between the ribs, as they have been pressed down, are a number of flattened, soft, punctulate bones, which are evidently the costal cartilages. Posteriorly four rows of them are seen, lying closely side by side, some of them eight or ten inches in length. The sternum, composed of the same material, has been so crushed and crumpled that its shape cannot be made out. The whole structure here, whether of ribs, cartilages or sternum, reminds one very strongly of such lizards as the Iguana or Monitor. There is no indication, however, in any specimen, of an episternum.

MEASUREMENTS OF RIBS.

Length, first thoracic rib, (chord)	200 millim.
Length, eleventh thoracic rib, (chord)	145
Length, thirteenth dorsal rib	68
Length, eighteenth dorsal rib	64
Length, thirty-fourth dorsal rib	52

The lengths of the different regions, as they lie in their natural relations, are as follows:

Skull	0.420 meters.	
Neck	0.225	
Trunk	1.360	
Tail	1.460	
Total	3.465	11 ft. 7 in.

The measurements of an excellent specimen of *C. tortor* are as follows:

Skull	0.630 meters.
Neck	0.360
Trunk, (thirty-three vertebrae preserved)	2.370

A very complete specimen of a *Liodon* in the Museum, in which the *complete* vertebral column is present, numbering one hundred and seventeen vertebrae, gives the following measurements. The skull is complete, save the most anterior portion.

Skull (approximated within narrow limits)	0.700 meters.	
Neck	0.430	
Trunk	1.760	
Tail	3.420	
Total	6.310	20 ft. 8 in.

The vertebral series in this specimen is composed of seven cervicals, twenty-three dorsals, seven pygials, and eighty chevron-caudals.

The relative proportions of the different regions in the two genera, as shown by the two specimens of *Clidastes* and *Liodon*, may be represented as follows. The first column is for *Clidastes*.

Skull	12.1	11.1
Neck	6.5	6.8
Trunk	39.2	28.0
Tail	42.3	54.1

Limbs.

The figures in plates II and III will give a sufficiently good idea of the limbs in this specimen. They are figured as they were lying, showing the outer sides of the coracoid, scapula and pelvic bones, and the palmar or plantar surface of the remaining bones.

Coracoid.

It will be observed in plates II and IV that there are two very different types of coracoid, one with a deep emargination, the other without the slightest indication of such. The same non-emarginate form occurs in *C. tortor*, as specimens in our Museum show, in *C. propython* Cope (Ext. Batr. etc. pl. xii, f. 16,) and in *C. dispar*, as figured by Marsh^[7], and as stated by

him in the same paper (“There is certainly no emargination in the coracoid of *Clidastes*, *Edestosaurus* and *Baptosaurus*, as specimens in the Yale Museum conclusively prove.”) It is true that Marsh in a later paper^[8] figured a specimen with emarginate coracoid under the name of *Edestosaurus dispar*, but it is certain that his identification of his own species was wrong, as will be seen by comparing his figures. From the senior author’s memory of the specimen with the emarginate coracoid figured, and from the figure itself he feels confident that the second specimen is *C. velox*.

That the emargination was overlooked by the author seems strange, as in the same paper in which this figure is given occurs the description of *Holosaurus*, *founded upon that very character*. If the emargination is sufficiently important to base a genus in the one case, then it should be in the other, and the character could not be applied to *Edestosaurus*, based upon characters which it hardly seems possible that the author himself could seriously consider, for *E. dispar* was the type of *Edestosaurus*.

It will be observed, further, that the figured coracoids differ very materially in size, those with the emargination pertaining to a small species, while *C. dispar* is one of the largest. In our Museum there are three specimens with the emarginate coracoid, all of them small or very small, the described specimen of *C. velox* being the largest.

The point of chief interest in this relation is the value that can be given to this character. Is it individual, specific or generic? Marsh has called it generic, but we think an examination of the two very complete specimens of *C. tortor* and *C. velox* in our Museum will convince any unprejudiced student that he is in error.

A comparison of the figures herewith given of the paddles will show their great resemblance, and these two forms of paddles have been figured because the species are the most unlike of any that we know in the genus. As all the small specimens seem to possess this character, and as they cannot be called immature specimens, we believe the character is a specific one. As Marsh says, typically both *Clidastes* and *Edestosaurus* have a non-emarginate coracoid, so that neither name could apply to the emarginate form, were it generically distinct.

Our Museum also contains both forms of the coracoid pertaining to the genus *Platecarpus*, of which *Holosaurus* is a synonym.

While studying the specimen above described, a striking similarity was observed to several other specimens already determined with confidence as *C. pumilus* Marsh. A more careful comparison failed to bring out any real differences beyond size, and even this was shown to be very inconstant.

The following comparison of the descriptions given by Marsh will be of interest.

C. pumilus.

C. velox.

TEETH. Nearly round at base Premaxillary and maxillary teeth somewhat curved and with smooth and subcompressed. smooth enamel.

QUADRATE. The rugose knob The great ala less curved than in *E.* near the distal end of the *dispar*, concave transversely on both quadrate is similar to that in *C.* surfaces. The alar process has its *Wymani* (just below the articular process very narrow in its posterior superior process is a extension over the great ala. No notch in prominent rugose knob with a posterior margin of external angle. On deep pit under it), but has no the ridge below the angle and nearly articular pit under it. The hook opposite the meatal pit is a strong is comparatively short and has a rugosity which is rudimentary or free compressed extremity. The wanting in *C. dispar*. The posterior articular margin is not deflected margin of the hook is only a narrow toward the meatus. tongue projecting towards the meatal pit, instead of a broad articular surface.

CERVICAL VERTEBRAE. Articular face transverse.
Articular face nearly vertical,
and having a broad transverse
outline with faint superior
emargination. The

hypapophysis stout and
transversely triangular.

The description, otherwise, shows no discrepancies of importance. The chief difference given by the author is the size, and this character we think our specimens show to be of little specific value. "It is a question of some importance how far difference in size among the Mosasauroids may be a test of difference in species. Among the numerous remains of these animals which have been discovered I have never yet observed any which presented any evidence relative to age. * * * In this view of the case, some of the many described species of Mosasauroids may have been founded on different sizes of the same."^[9]

The length of the cervical vertebrae in the specimen above described is thirty-seven or thirty-eight millimeters. The cervical vertebrae in two specimens referred to *C. pumilus* have lengths respectively of twenty-two and thirty millimeters. In the type specimen of *C. velox* they must have had a length of at least forty-two millimeters.

It thus appears that, between the smallest specimen, which, in life, could have hardly exceeded eight feet in length, our specimens, indistinguishable anatomically, represent forms of ten and twelve feet, while the type itself was about fifteen feet in length.

Of the material originally referred to *C. pumilus*, there are in the collection five or more specimens, which, altogether, furnish nearly every part of the skeleton. They present no tangible differences from the skeleton of *C. velox* described above. There can be, hence, little or no doubt but that the name *C. pumilus* is a synonym.

It is hardly possible to say with certainty that *C. affinis* Leidy is or is not the same as *C. velox*, but, so far as the description goes, we can find few differences. The type is of about the same size as the type of *C. velox*, and the figures agree well with the bones of the skeleton described. Although the description was not published till 1873, the author makes no mention of the species of Marsh's. Leidy describes the back teeth as having the enamel strongly striated, with the surface presenting evidences of subdivision into narrow planes. In this respect, only, it disagrees with the specimen.

Plioplatecarpus Dollo is described by its author as having a sacrum of two conjoined vertebrae,^[10] by reason of which it is placed in a separate family from the rest of the *Pythonomorpha*. It may be presumptuous to express a doubt of the genuineness of the sacrum, and yet, save from the fact that the author found two specimens quite alike, one might doubt it strongly. It is not very rare that two, or even three vertebrae are found united from injury in these animals, and such would readily account for the consolidation as figured and described by Dollo, except for the coincidence of the second specimen. A stronger reason for doubt is the statement that the consolidated vertebrae belong to the posterior “lumbar” region, and that the last vertebrae had small tubercles indicative of chevrons. In the reptiles which we have examined, the chevrons do not begin immediately behind the pelvis, but are separated by a longer or shorter region in which the vertebrae bear elongated diapophyses alone. If the conjoined vertebrae figured by Dollo are in reality sacral, it would appear that the animal is an exception to *Clidastes* and such lizards as we have examined. Furthermore, the pelvis must have been of a different structure from that in the Kansas genera of the *Pythonomorpha*, for, in these, it is evident that the ilium had an oblique position, and could have been attached to but a single diapophysis.

CLIDASTES WESTII, N. SP.

A specimen of much interest in the University collection differs so markedly from the other forms represented by specimens, as also from the descriptions of the known species, that we are constrained to regard it as new. It was collected by Mr. C. H. Sternberg from the uppermost of the Niobrara beds, in the vicinity of the old town of Sheridan. The character of the associated invertebrate fossils seems to indicate a different geological horizon, either the Fox Hills group, or transition beds to that group. The specimen consists of a complete lower jaw, quadrate, portions of the skull, the larger part of the vertebral column, and the incomplete hind and fore paddles. The vertebrae preserved are in two series, the one, numbering thirty-three, continuous with the skull; the other, sixty-three in number, all chevron caudals. The terminal caudals preserved indicate that there were several more in life, perhaps five or ten; the first of the series was evidently among the first of those which bore chevrons. Altogether the tail may have

had seventy-five chevron caudals. The lengths of the two series are respectively seventy-one and seventy-two inches. Assuming that there was the same number of precaudal vertebrae as in *C. velox*, the entire vertebral column would have measured in life fifteen feet and four inches. The lower jaw shows the skull to have been very nearly twenty-four inches in length, making, for the animal when alive, a length of seventeen and one-half feet. This is one of the largest species, and it is interesting to observe that the real size here, as usually elsewhere among fossil vertebrates, is less than supposed. It is doubtful whether there is a *Clidastes* known that exceeded twenty feet in length.

While the skeleton was only about one half longer than the specimen of *C. velox* described in the foregoing pages, or of about the same length as a very complete specimen of *C. tortor* in the museum, the proportions of the animal were very much stouter. The figures given in plate VI of the twenty-fifth, or eighteenth dorsal, vertebra will show the relations between length and breadth: it is upon these remarkably stout proportions, and the shape of the articular faces, as indicated by the figures and by the measurements appended, that the species is chiefly based. The articular surfaces of the basal caudal vertebrae are remarkably triangular in shape, with the angles rounded, and the sides of nearly equal length. This triangular shape is persistent for the first twenty of the series as they are preserved. The paddles, as shown in plates IV and V, show much stouter proportions than in either *C. velox* or *C. tortor*.

The species comes nearest to *C. stenops* Cope, but it seems hardly the same. It is, also, evidently allied to *C. dispar* Marsh. From these and other described species, the following, extracted from the original descriptions, will serve to show the differences, in comparison with the specimen of *C. Westii*.

C. dispar.

The articular faces in the cervicals are a broad transverse oval, faintly emarginated above for the neural canal. In the dorsals and lumbar the cup continues transverse, and the emargination is deeper, but in the anterior caudals the outline becomes a vertical oval. There appears to have been thirteen mandibular teeth.

Length of axis with odontoid process	32 lines	100
Width between diapophyses	26.8	103
Length from edge of cup to end of ball in eleventh vertebra	25	100
Width of ball	14	56
Depth of ball	12	43

C. Wymani.

In the cervical vertebrae, the outline of the articular faces is transversely cordate. The centra of the anterior dorsals are elongate, and much constricted behind the diapophyses. In the anterior caudals, the articular faces are a broad vertical oval.

Length of axis with odontoid process	19 lines	100
Width between diapophyses	17	89.4
Width of ball	8	42.1
Depth of ball	7	36.7
Length of sixth cervical, without ball	13	100
Width of cup	9	69.1

C. rex.

The cervical vertebrae have very broad, transversely oval faces, with indications of emargination. The dorsals are elongated, with transverse faces, and a distinct superior excavation for neural canal. The articular ends of the anterior caudals are vertically oval.

Length of posterior cervical vertebrae	44 mm	100
Vertical diameter of ball	24	54.5
Transverse diameter	29.5	67
Length of a dorsal vertebra	52	

C. stenops.

The anterior caudals possess wide diapophyses. Their articular faces are a vertical oval, a little contracted above, sometimes a straight outline. They

present a peculiarly elongate form.

Length of axis (alone)	60 mm	100
Vertical diameter of ball	27	45
Transverse diameter of ball	27	45
Length of the mandible	720	100
Depth at coronoid process	150	20.9

MEASUREMENTS OF CLIDASTES WESTII.

Length of dentary	400 millim.
Depth opposite the first tooth	20
Depth opposite last tooth	62
Entire extent of mandible	630
Greatest depth at coronoid process	95
2. Length of axis with odontoid process	80
Length of axis without odontoid process	70
Vertical diameter of ball	24
Transverse diameter of ball	33
4. Length of fourth cervical vertebra to rim of ball	49
Expanse of diapophyses	82
5. Length of fifth cervical to rim of ball	49
Transverse diameter of ball	35
Vertical diameter of ball	28
Expanse of diapophyses	90
8. Length of eighth vertebra to rim of ball	53
Expanse of diapophyses	90
14. Length to rim of ball	54
Transverse diameter of ball	40
Vertical diameter of ball	33
Expanse of diapophyses	100
18. Length to rim of ball	50

Transverse diameter of ball	40
Vertical diameter of ball	36
Expanse of diapophyses	100
23. Length to rim of ball	50
Transverse diameter of ball	41
Expanse of diapophyses	100
25. Length to rim of ball	52
Transverse diameter of ball	43
Vertical diameter of ball	43
Expanse of diapophyses	100
30. Length to rim of ball	54
Transverse diameter of ball	46

This species is named in memory of Judge E. P. West, lately deceased, to whom our Museum owes so much for his long, diligent and faithful labors in the collection and preparation of the geological material.

[ERRATUM](#): P. 17, line 15, for “*Edestosaurus*,” read *Clidastes*, and in next line, strike out “Proc. Acad.” etc.

Notes and Descriptions of Syrphidae.

BY W. A. SNOW.

WITH [PLATE VII.](#)

Among the insects obtained by Prof. F. H. Snow in a recent trip to Colorado, is an excellent representative collection of the Diptera. The material for the following notes on Syrphidae is chiefly drawn from this collection. That such a collection affords so many points of interest in this, one of the best studied families of North American Diptera, is an evidence of the rich field that is presented by this important and little-studied order of insects.

CALLICERA.

Callicera Panzer, Fauna Germanica, 1806.

Callicera is a small genus hitherto supposed to be peculiar to Europe. The species are found in the high mountains, where the males are often taken while hovering in the air. The present collection includes numerous specimens of a species taken near the summit of Mt. Deception, in Manitou Park, Colorado, at an altitude of nine thousand feet.

The occurrence of members of this genus in the western part of the United States is a fact of especial interest and further substantiates the rule that American forms common to Europe are more apt to occur in the western regions. *Arctophila flagrans* Osten Sacken, is a case precisely similar to the present one, belonging as it does to a small European genus of mountain flies, and described from Colorado.

As the genus is a new one to our fauna, I here give an amended transcription of the generic characters from Schiner's Fauna Austriaca, to include the new species, which differs only in unimportant details.

Callicera.

Rather large, stout, green or black species with metallic lustre and abundant, long pile. Head hemispherical, somewhat broader than the thorax. Antennae porrect, longer than the head, somewhat remote at their base, inserted upon a protuberance of the front; first joint sometimes elongate; second joint shorter than, or as long as, the first joint; third joint one to three times the length of the first two joints taken together, with a short, terminal style. Face broad, under the antennae concave in profile; an obtuse tubercle below the middle; on the sides thickly covered with pile. Proboscis rather prominent, with broad labella. Eyes hairy, holoptic in the male. Abdomen elliptical, as long or longer than the thorax. Legs moderately strong. Third longitudinal vein straight, first posterior cell distally short petiolate; marginal cell open; cross-vein situated near the middle of the discal cell, oblique.

Callicera montensis, n. sp.,

Plate vii, f. 4.

MALE. Black, densely golden red pilose. Frontal triangle, face and cheeks deep black, shining, covered thickly with black pile, save a median facial stripe. Antennae black, basal third of third joint on the under side red; first joint short; second joint not more than half as long as the first; third joint three times as long as the first and second joints taken together; gradually broadened for a third of its length, and then attenuated; style white. Eyes thickly clothed with golden pile. Thorax and abdomen covered everywhere with long golden red pile. Legs black; tarsal joints below and at their articulations reddish. Wings nearly hyaline, brownish on the anterior basal portion; stigma yellow.

Length 11 millimeters. Three specimens, Colorado.

The genus may be distinguished from *Pelecocera*, in Williston's dichotomic table of the genera of North American Syrphidae, by the pilose eyes.

Microdon megalogaster, n. sp.,

Plate vii, f. 1.

MALE. Large, yellowish pilose species, in shape globose. Antennae reddish black, the first joint about as long as the following two together; second joint not one-third as long as the third. Face dark metallic green, shining, thickly covered with golden yellow pile. Front black, with similar pile, narrowed in the middle. Eyes bare. Thorax and scutellum deep metallic green, with long, thick, golden pile; scutellum gently emarginate, the small obtuse tubercles approximate. Abdomen short and broad, black, moderately shining; first two segments and the hypopygium somewhat green; pile at base yellow, elsewhere short, black. Legs black, with black pile; front tibiae and their metatarsi, on the inner side, with short golden pile; hind metatarsi incrassate and longer than the three following joints taken together. Wings uniformly subinfusate; veins at the outer part of the first posterior and discal cells sinuous and rounded.

Length 12 millimeters. One specimen.

Chrysotoxum derivatum Walker.

Eight specimens from Colorado, which vary not a little from each other and from Williston's description. They seem to belong here, however, better than elsewhere. In one specimen, the second joint of the antennae is shorter than the first, and only one-fourth the length of the third. In five examples the second abdominal cross-band is not interrupted; in the others it is distinctly parted. In two, the third band does not reach the yellow of the broad hind margin; in two others it barely touches it; in five, the two bands broadly coalesce. The yellow of the fifth segment, in four specimens, incloses a black, inverted V; in two others an inverted Y.

Paragus bicolor Fabr.

Three specimens, Colorado. These may be located under Schiner's variety *taeniatus*.

Melanostoma stegnum Say.

Syrphus stegnus Say, Journ. Acad. Phil. vi, p. 163.

Melanostoma tigrina Osten Sacken, Western Diptera, p. 323.

Melanostoma stegnum Williston, Biol. Centr.-Amer. Diptera, iii, p. 10.

Eleven specimens, Colorado, which answer well to the descriptions. The metallic band of the fourth abdominal segment is sometimes interrupted, and there is usually a triangular opaque black spot near the anterior border of the fifth segment. "The female, hitherto unknown, has the front broad above, pollinose, except on the upper part, and with black pile; the thorax more shining metallic blue; the tibiae yellow, and on the third and fourth abdominal segments there is a narrow shining stripe, bisecting the black, as in the fourth segment of the male. The male has some long black hairs on the outer side of the front and middle tibiae, which are inconspicuous in the female. It is evident, from the lighter color of the tibiae, that Say's specimens were females." Williston, l. c.

Melanostoma mellinum Linne.

A single female specimen from Manitou Park.

Melanostoma, n. sp. ?

MALE. Face and front dark metallic blue, shining, thinly covered with light-colored pollen; tubercle and epistoma black, shining, the former small. Antennae black, third joint yellowish red below, oblong. Pile of frontal and vertical triangles dusky. Thorax bronze-black, shining, sometimes bluish black, the pubescence white. Halteres yellowish. Abdomen long and narrow, with almost parallel sides; first segment metallic blue, shining; second segment opaque, or subopaque, black, with a light metallescent scallop on the sides, reaching to the distal third of the segment; third and fourth segments similar, marked anteriorly by a wide, interrupted, or subinterrupted blue fascia, deeply and widely emarginated, or concave behind; hind border of the third, and sometimes of the second segment, narrowly brown; fifth segment and the hypopygium metallic bluish green; sides of the abdomen with silvery white pile, longest and thickest at the base; the blue marking are whitish pruinose. Femora, except the tip, a broad ring on the tibiae, and the four posterior tarsi, black; elsewhere brownish or yellowish. Wings hyaline, stigma yellowish.

Length 7-8 millimeters.

Eupeodes volucris, Osten Sacken.

Numerous specimens, Colorado.

Syrphus arcuatus Fallen.

Four specimens, Colorado. These specimens vary not a little from each other, and somewhat from the descriptions. One female is very small, not over seven millimeters in length, and with the spots on the third and fourth abdominal segments hardly oblique. One male has the hind femora black as far as the tip, while in three females the black does not extend beyond the middle.

Syrphus disjectus Williston.

A single female specimen, from Colorado, agrees well with the description drawn from males. The pile of the thorax is more whitish than orange-yellow, and there are light colored lateral margins on the anterior part of the thorax.

Syrphus ruficauda, n. sp.,

Plate vii, f. 3.

MALE. Eyes bare. Face greenish yellow on the sides, yellow in the middle; a rather broad black line marks the border of the mouth and is lost in the black of the cheeks. Frontal triangle yellow, with long black pile. Antennae dark brown, more or less reddish below. Pile of occiput light yellow. Dorsum of thorax deep metallic green, the scutellum olivaceous yellow; both with light yellow pile. First segment of the abdomen shining black; second segment opaque black, with the lateral margins and hind border shining, and with a broad, yellow, interrupted band, not reaching the lateral margins; third segment similar, but with the yellow band somewhat wider, interrupted or subinterrupted and slightly bilaterally oblique; fourth and fifth segments orange-red, the sides narrowly black; the fourth segment shows indistinctly a broad interrupted band of a somewhat lighter color, corresponding to the yellow bands of the preceding segments. Legs light brown; basal third of the front and middle femora and basal half of the hind femora black. Wings hyaline, stigma yellowish.

FEMALE. Head wanting. Thorax purplish brown. The yellow band on the second abdominal segment narrower, the second band straight, narrower and interrupted. Legs light brown, except the proximal end of the femora, which is black.

Length 9 millimeters. Three males and one female, Colorado.

Syrphus pauxillus Williston.

Two specimens from Colorado undoubtedly come here. The species was described from a single male specimen. A female specimen offers the following differences or additions: Length nine millimeters, mesonotum more greenish black or bronze, the pile obscure whitish; fifth abdominal segment without yellow spots on the anterior angles; legs yellow, with the basal half of the front and middle femora, the hind femora except the tip, a broad band on the hind tibiae, and the hind tarsi, black.

Syrphus ribesii Linne.

Five specimens, Colorado.

Syrphus americanus Wiedemann.

Numerous specimens, Colorado.

Syrphus umbellatarum Schiner.

Five female specimens, Colorado. The only western locality heretofore given is Arizona (Williston).

Allograpta obliqua Say.

Five specimens, Colorado.

Mesogramma marginatum Say.

Numerous specimens from Colorado, showing very great variation.

Sphaerophoria cylindrica Say.

Twenty specimens, Colorado. I think the specimens belong here, though a positive identification is hardly possible at present.

Rhingia nasica Say.

One specimen, Colorado. This is the first time that this species has been recorded from beyond the Mississippi.

Copestylum marginatum Say.

Two specimens, Colorado, representing the extremes of variation in the species. The male corresponds to *C. lentum* Williston. Specimens of this species were bred from *Opuntia missouriensis*, in company with others of *Volucella fasciata* Macq.

Sericomyia militaris Walker.

Sixteen specimens from Minnesota and Colorado vary in the markings of the second abdominal segment, and in the color of the legs. Some have no spots at all on the second segment; in others the two yellow dots are conspicuous, approaching, in size and shape, the markings of the third segment. The tibiae vary from light yellow to reddish brown.

Brachyopa cynops, n. sp.,

Plate vii, f. 2.

Head light yellowish brown, largely concealed beneath light glistening pollen; the shining ground color shows just above the antennae and in a stripe on the cheeks, extending from the eye to the mouth opening. Antennae wanting. Dorsum of thorax brown, covered with grayish pollen; anteriorly with two approximated, linear, blackish stripes; laterally with a broad, interrupted stripe. Scutellum light brown, with yellowish pollen. Abdomen but little longer than broad; yellowish gray pollinose; second segment with a circular brown spot in the anterior corners; the two following segments are marked with corresponding elliptical spots, and, in the middle of the anterior border with a triangular spot; on the fifth segment are two small round spots. Legs uniformly reddish brown, with light colored pollen and short whitish pile. Wing hyaline, distinctly clouded at anterior cross-vein, on the veins at the anterior outer corner of the discal cell and on the ultimate section of the fourth vein; posterior cross-vein about as long as the penultimate section of the fourth vein, the included angle obtuse.

Length 5 millimeters. One specimen, Colorado.

Eristalis latifrons Loew.

Numerous specimens, Colorado. The commonest Syrphid of the mountain meadows. Some specimens have very indistinct brownish spots on the second abdominal segment, and, when this is the case, the middle of the wing generally shows a brown spot, and brown clouds along the anterior veins between the spot and the base of the wing.

***Eristalis brousi* Williston.**

One male specimen, Colorado.

***Helophilus latifrons* Loew.**

Numerous specimens, Colorado.

***Xylota flavitibia* Bigot.**

Eight specimens, Colorado. The glistening pile of the face and front varies from white to a golden yellow. On the dorsum of the thorax purplish stripes are distinctly visible. The fourth segment of the male abdomen is often red, as in the female abdomen.

***Syritta pipiens* Linne.**

Eight specimens, Colorado.

***Criorrhina umbratilis* Williston.**

A single, male specimen, collected by Mr. W. J. Coleman, at Lawrence, and agreeing exactly with the description. The only other known specimen of this species is the type, at Washington, from Connecticut.

***Spilomyia quadrifasciata* Say.**

Seven specimens, Lawrence, Kansas, (F. H. Snow and E. S. Tucker). The species has not hitherto been recorded west of New York.

Notes on *Melitera Dentata* Grote.

BY VERNON L. KELLOGG.

WITH [PLATE VIII.](#)

At the meeting of the Entomological Club of the A. A. A. S., held in August, 1891, at Washington, Dr. Riley called attention to the habits of *Melitera prodenialis* Walker. The larvae burrow into and feed upon the fleshy leaves of the prickly pear, *Opuntia*. Dr. Riley's specimens came from Florida. Prof. J. B. Smith has recently bred the moth from the prickly pear in New Jersey. His notes were presented at the same meeting of the Club, and the brief references to the interesting notes of Doctors Riley and Smith, made in the Canadian Entomologist (v. xxiii, num. 11, pp. 242 and 256), suggest the presentation of the following notes on *Melitera dentata* Grote, the western species of this Phycitid genus.

Chancellor F. H. Snow, of this University, while investigating a grasshopper "outbreak" (*Dissosteira longipennis*) in eastern Colorado in July, 1891, noted the withered and dying condition of many leaves of the common prickly pear cactus (*Opuntia missouriensis*), and on examining the leaves found in them certain large, naked, bluish larvae. The larvae were imbedded in the fleshy leaves, eating away the soft inner tissue. The hollowed-out spaces were nearly filled with irregularly spherical, yellowish, translucent casts. The attacked leaves were withered and brown without. Prof. Snow took a few leaves and larvae on July 16, near Arriba, Colorado, and brought them to the laboratory.

The larvae were put into breeding-cage on July 18. On July 28 one larva had spun up and pupated in a corner of the cage behind a small porcelain dish. Another had made a cocoon in a broken, empty pupa-case of *Eacles imperialis*, but died before pupating. On August — the adults appeared, and have been determined by Prof. J. B. Smith as *M. dentata*, Grote. As I

am aware of no description of the earlier stages of this species, I record the following notes of description:

EGG. About 1-1.2 millimeters in diameter, surface with broad, meridian-like furrows from one pole for about one-third of the distance to the other pole. Color, creamy white.

LARVA. Food plant, *Opuntia missouriensis*, prickly pear cactus, burrowing into the fleshy leaves and eating the soft, succulent, inner tissues. Length, 40 millimeters. Five pairs of prolegs. Color, one specimen, ultramarine blue; skin, semi-transparent and shining anteriorly, dead blue on dorsum; second specimen, buffy with a bluish suffusion, blue between segments, prolegs bluish, and last abdominal segment blue, especially below; skin more opaque than in first specimen. No pronounced markings of skin; spiracles shining black and present on first thoracic and first to tenth abdominal segments. Head flattened, slightly narrower than first thoracic segment, umber. Prothoracic shield well marked, brownish black; anal shield, smoky brownish. Clothing, limited to tubercled hairs sparsely distributed as follows: a subdorsal line of small tubercles, two tubercles to a segment, each tubercle bearing three short, fine hairs; a supra-stigmatic line, one tubercle to each segment, each tubercle bearing three to four fine hairs; a similar infra-stigmatic line; a sub-ventral line of tubercles, bearing usually four fine hairs, the tubercles of the three thoracic segments in this line situated at base of legs outside, and similarly as to the prolegs on the third to sixth abdominal segments. The tubercles in all the lines are faintly smoky. The larva is rather heavy, and rotund in form, tapering toward both head and posterior segment. It moves with a lumbering gait, but rather rapidly.

CHRYsalis. Length, 20 millimeters; in cocoon of silk, loosely covered with small dirt-masses. As made in the breeding cage the cocoons were above ground, but concealed under or in available objects.

ADULT. The adults obtained from the breeding cage, (there are no others in our collection), are easily distinguished from *prodenialis* Wlk., by the much stronger dentations of the outer line of the primaries. Prof. Smith kindly sent a specimen of *prodenialis* taken at Ocean Grove, New Jersey, for comparison. The row of marginal black spots on the primaries which Hulst (Tran. Am. Ent. Soc., v. xvii, p. 172) mentions as distinctive of

dentata is as pronounced in Prof. Smith's specimen of *prodenialis* as in our *dentata*. The much lighter color of the primaries, head and thorax in *dentata* as mentioned by Hulst is characteristic. An interesting feature in the venation of the hind wings in our bred specimens of *dentata* is the considerable coalescence of the sub-costal and costal veins. Vein five is wanting, as mentioned by Hulst. In addition, there is further departure from a normal venation, in that vein seven after rising with six from its stem, (Hulst says: "Six short stemmed with seven"), coalesces for a short distance with eight and then runs free to the margin. Behind the forking of seven and six the stem (remnant of sub-costal) unites with the costal, and its basal portion is wholly merged with the forward vein. This partial disappearance of the sub-costal seems to be shared by *prodenialis* and is probably characteristic of the genus.

Prof. Smith, as recorded in the Canadian Naturalist, v. viii, p. 242, (1891), bred several specimens of *Volucella fasciata*, a Syrphid fly, from the same prickly pear leaves in which the *Melitera* larvae were living. It is interesting to note that pupariae and later, adults of *Volucella fasciata* and *Copestylum marginatum*, a closely allied Syrphid, were noted in the Opuntia leaves from which *M. dentata* was bred. (See note by Dr. Williston, Entomological News, v. ii, p. 165, 1891).

Diptera Brasiliana.

BY S. W. WILLISTON.

PART II. [\[11\]](#)

CONOPS.

- | | |
|---|-------------------------------|
| 1. First basal cell hyaline | 2 |
| First basal cell clouded throughout | 6 |
| 2. Third joint of the antennae as long as the first two together; | |
| small species | <i>parvus</i> , n. sp. |
| Third joint of the antennae but little if any longer than the | |
| second joint | 3 |
| 3. First posterior cell hyaline | 4 |
| First posterior cell more or less clouded | 5 |
| 4. Cheeks yellow | <i>angustifrons</i> , n. sp. |
| Cheeks black | <i>ornatus</i> , n. sp. |
| 5. Face black in ground-color | <i>argentifacies</i> , n. sp. |
| Face yellow, large species | <i>grandis</i> , n. sp. |
| 6. Red species; front red | <i>rufus</i> , n. sp. |
| Black species; front black | 7 |
| 7. Face and cheeks black in ground-color | <i>magnus</i> , n. sp. |
| Face and cheeks yellow | <i>inornatus</i> , n. sp. |

1. *Conops magnus*, n. sp.

FEMALE. Front black, shining, the vertical callosity somewhat reddish. Face and cheeks yellowish brown, the orbits silvery pollinose. Antennae brownish black; second and third joints subequal, first joint about two-thirds the length of the second; third joint of the style with a long bristly extremity. Thorax shining black; pleurae lightly whitish pollinose. Abdomen deep black, opaque; lightly whitish pollinose posteriorly; ventral process of the fifth segment large. Wings deep brown in front, extending through the two basal cells, and the basal part of the discal cell; outer part of the first posterior cell subhyaline, as also behind the streak corresponding to the spurious vein of the Syrphidae. Legs black; base of the femora, of the tibiae, and of the tarsi, somewhat yellowish.

Length 21-24 millimeters. Six specimens, Chapada, H. H. Smith.

2. *Conops grandis*, n. sp.

FEMALE. Front black, the lower margin of the vertical callosity reddish; just below the callosity opaque, elsewhere shining. Antennae black; the second and third joints of nearly equal length; the first joint about two-thirds the length of the second joint; style with a long bristly extremity. Face and cheeks light yellow, the orbital margins of the former silvery or light golden pollinose. Thorax black, the mesonotum shining, the pleurae lightly whitish pollinose. Abdomen deep black; posteriorly lightly pollinose. Wings brown in front; first posterior cell and the space behind the streak corresponding to the spurious vein of the Syrphidae in the first posterior cell, pure hyaline; outer part of the first posterior cell subhyaline; a brown streak in front of the fifth vein. Legs black; the tibiae and basal joints of the tarsi in large part reddish or yellowish; pulvilli light yellow; ventral process of the fifth segment extraordinarily large; seventh segment as long as the three preceding together.

MALE. Abdomen in ground-color black, either wholly so, or more or less, or rarely entirely, red; the ground color, save at the base, however, is almost wholly obscured by reddish brown pollen.

Length 19-23 millimeters. Six specimens, Chapada, H. H. Smith.

3. *Conops rufus*, n. sp.

MALE, FEMALE. Head red; face in the depression yellow, on the sides with a silvery sheen. Antennae black; first joint red, more than half of the length of the second joint; second joint sometimes reddish at the base; third joint about as long as the second joint, stout; third joint of the style suddenly attenuated into a moderately long bristly extremity. Thorax red; mesonotum with a median black stripe, and an oval, more or less distinct spot on either side; a golden pollinose spot on the inner side of each humerus. Abdomen red, lightly pollinose, the median segments more or less black; ventral process in the female large; the sixth segment in the same sex about as long as the two preceding together. Legs red, the tarsi a little darker, the pulvilli and the ungues, save their black tip, yellow. Wings brown in front, the brown extending to the fifth vein in the basal part of the discal cell; the space behind the spurious vein in the first posterior cell hyaline; the outer part of the same cell subhyaline.

Length 16-17 millimeters. Two specimens, Chapada, H. H. Smith.

4. *Conops angustifrons*, n. sp.

MALE. Front much longer than wide; black, shining at the vertex and below; an opaque band below the vertical callosity. Antennae black, the third joint somewhat reddish below towards the base; the first joint about half of the length of the third joint; third joint distinctly shorter than the second, rather broad at the base; style small, attenuate. Face, cheeks and the lower part of the occiput wholly light yellow. Thorax opaque black; a whitish pollinose spot on the inner side of each humerus; vertical pleural pollinose spot not distinctly limited above; a row of dorso-pleural, at least two prescutellar, and four scutellar, well-developed bristles. Abdomen subopaque black; second segment yellow at the base; sixth segment opaque golden yellow pollinose. Wings brownish before the third longitudinal vein, the first basal and the first posterior cells wholly hyaline; a streak before the fifth vein. Legs deep brown; the base of all the tibiae, the large pulvilli, and the claws (except their tips) yellow.

Length 12 millimeters. One specimen, Chapada, H. H. Smith. This species is peculiar in its narrow front, bristles of the thorax, and hyaline first posterior cell.

5. *Conops nobilis*, n. sp.

FEMALE. Head black; front, below the vertical callosity, except a crescentic space above the base of the antennae, opaque; face, on the sides and in the depression, with a conspicuous, light yellowish silvery reflection; in an oblique light from above the ground-color wholly concealed. Antennae black; the reddish first joint about two-thirds the length of the third joint; the third joint about two-thirds of the length of the slender second joint; third joint of the style with a short bristly extremity. Thorax black, lightly pollinose, opaque; on the front margin, and near the humeri, velvety; in the middle in front distinctly whitish when seen from behind. Abdomen black, subshining; second segment deep opaque black, save on the anterior part, where it is whitish pollinose; ventral process of the fifth segment small. Legs black; the tarsi and claws (save their extreme tips) light yellow; pulvilli very large, yellow; the tarsi dilated. Wings unequally brown in front, scarcely extending beyond the third vein, save in the first posterior cell; the costal cell and the outer part of the wing in front of the third vein of a lighter color.

Length 12 millimeters. One specimen, Chapada, H. H. Smith.

6. *Conops inornatus*, n. sp.

MALE. Front black, shining, the vertical callosity reddish. Face yellow, with golden pollen on the sides extending up on the lower part of the front. Cheeks wholly yellow. Thorax black, shining, lightly pollinose; margins of the thorax and of the scutellum with moderately large bristles. Abdomen slender, black, shining; the narrow hind margins of the third and fourth segments, the fifth on the sides and behind, and the sixth nearly wholly, light golden pollinose. Legs brown; base of tibiae yellow; basal joints of the tarsi yellowish. Wings subhyaline, without distinct picture, though the color is more intense in front; yellow in the costal cell.

FEMALE. Wings distinctly brown before the third vein and in the basal cells and proximal portion of the discal cell. Abdomen diffusely whitish pollinose behind; the second segment largely reddish; ventral process of the fifth segment small.

Length 10 millimeters. Two specimens, Chapada, H. H. Smith.

7. *Conops ornatus*, n. sp.

MALE. Vertical callosity reddish; below it an opaque black band, connected in the middle with a V-shaped spot about the base of the antennae; the front elsewhere, and the face for the greater part, light yellow, the sides of the latter with a broad silvery sheen. Cheeks black. Antennae red; the first joint a little shorter than the third joint; second joint about twice the length of the first; style short, thick. Thorax black, opaque; near the humeri and behind, as also on the scutellum, thickly golden pollinose; pleurae diffusely pollinose. Abdomen opaque black; the hind margin of the first three segments, and the remainder of the abdomen, save spots on the sides of the fourth and fifth segments, thickly light golden pollinose. Legs reddish brown, the base of the tibiae and the basal joints of the tarsi yellowish. The brown of the wings extends to the third vein and through the middle of the first posterior cell; costal and subcostal cells lighter colored.

Length 11 millimeters. Two specimens, Chapada, H. H. Smith.

8. *Conops parvus*, n. sp.

FEMALE. Closely allied to *C. sylvosus* Williston, but differs in the lighter colored antennae and their more elongated third joint, which is as long as the first two joints together; in the wings being wholly grayish hyaline, save a quadrate brown spot in front a little beyond the middle; and in the lighter colored legs and abdomen. The proboscis is as long as the antennae; the legs are brown or brownish yellow.

Length 8 millimeters. Two specimens, Chapada, H. H. Smith.

Explanation of Plates.

[PLATE I.](#) Skull of *Pteranodon* sp., one-fifth natural size.

[PLATE II.](#) Left front paddle of *Clidastes velox* Marsh, two-thirds natural size. *C*, coracoid; *S*, scapula; *H*, humerus; *I*, first digit; *V*, fifth digit.

[PLATE III.](#) Left hind paddle of *Clidastes velox* Marsh, two-thirds natural size. *Il*, ilium; *P*, pubis; *Is*, ischium; *F*, femur; *T*, tibia; *Fb*, fibula; *I*, first metatarsal.

PLATE IV. Right front paddle of *Clidastes Westii* Williston, one-third natural size. *S*, scapula; *C*, coracoid; *H*, humerus; *R*, radius; *U*, ulna; *I, IV*, first, fourth digits.

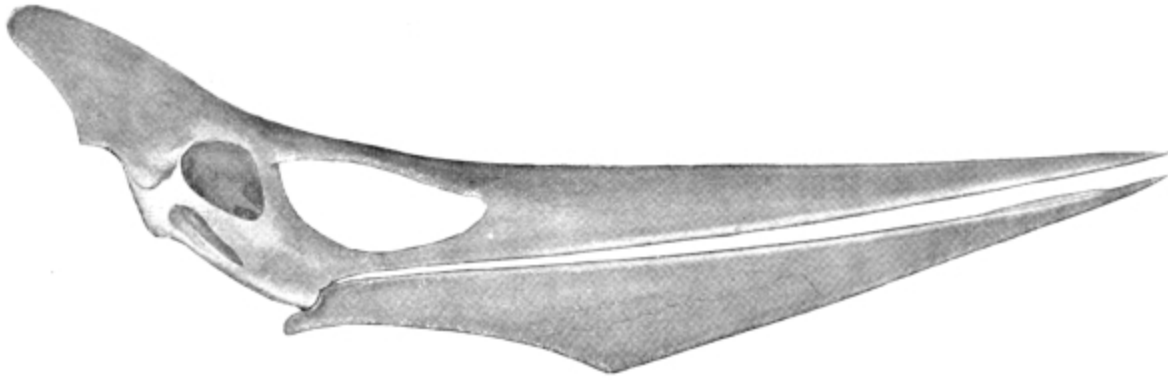
PLATE V. Right hind paddle of *Clidastes Westii* Williston, one-half natural size.

PLATE VI. Eighteenth dorsal vertebra of *Clidastes Westii* Williston, natural size. Fig. 1, centrum from behind; fig. 2, from below.

PLATE VII. Fig. 1, *Microdon megalogaster* Snow; fig. 2, *Brachyopa cynops* Snow; fig. 3, *Syrphus ruficauda* Snow; fig. 4, *Callicera montensis* Snow; fig. 5, *Tropidomyia bimaculata* Williston; fig. 6, *Rhingiopsis rostrata* Roeder; fig. 7, *Ancanthina hieroglyphica* Wiedemann.

PLATE VIII. *Melitera dentata*. Adult, silken cocoon and outer layer of dirt-masses held together by silken threads; larva (shaded); larva in outline showing position and number of tubercled hairs; hind wing of adult showing venation.

PLATE I. Skull of *Pteranodon* sp., one-fifth natural size.



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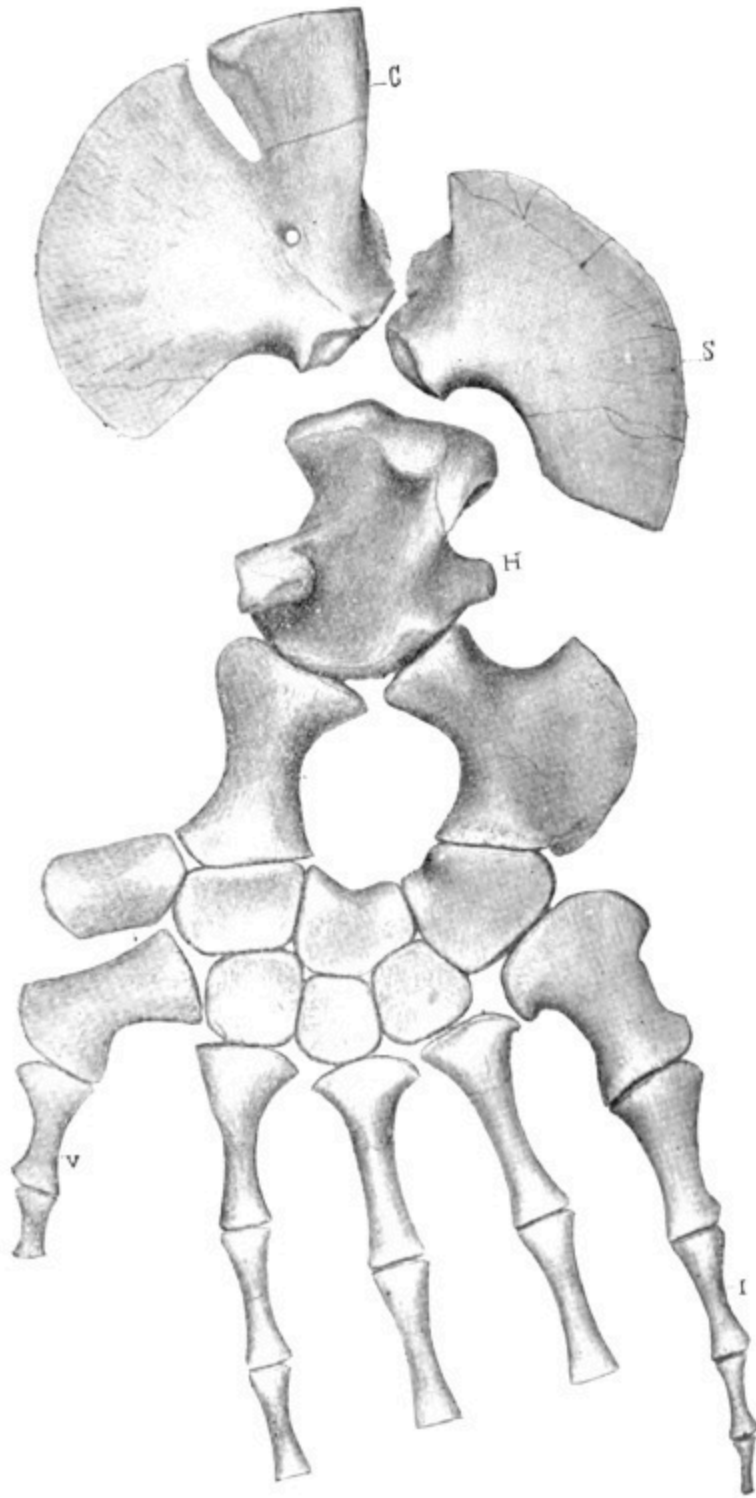
S. W. Williston.



PLATE II.

Left front paddle of *Clidastes velox* Marsh, two-thirds natural size. *C*, coracoid; *S*, scapula; *H*, humerus; *I*, first digit; *V*, fifth digit.

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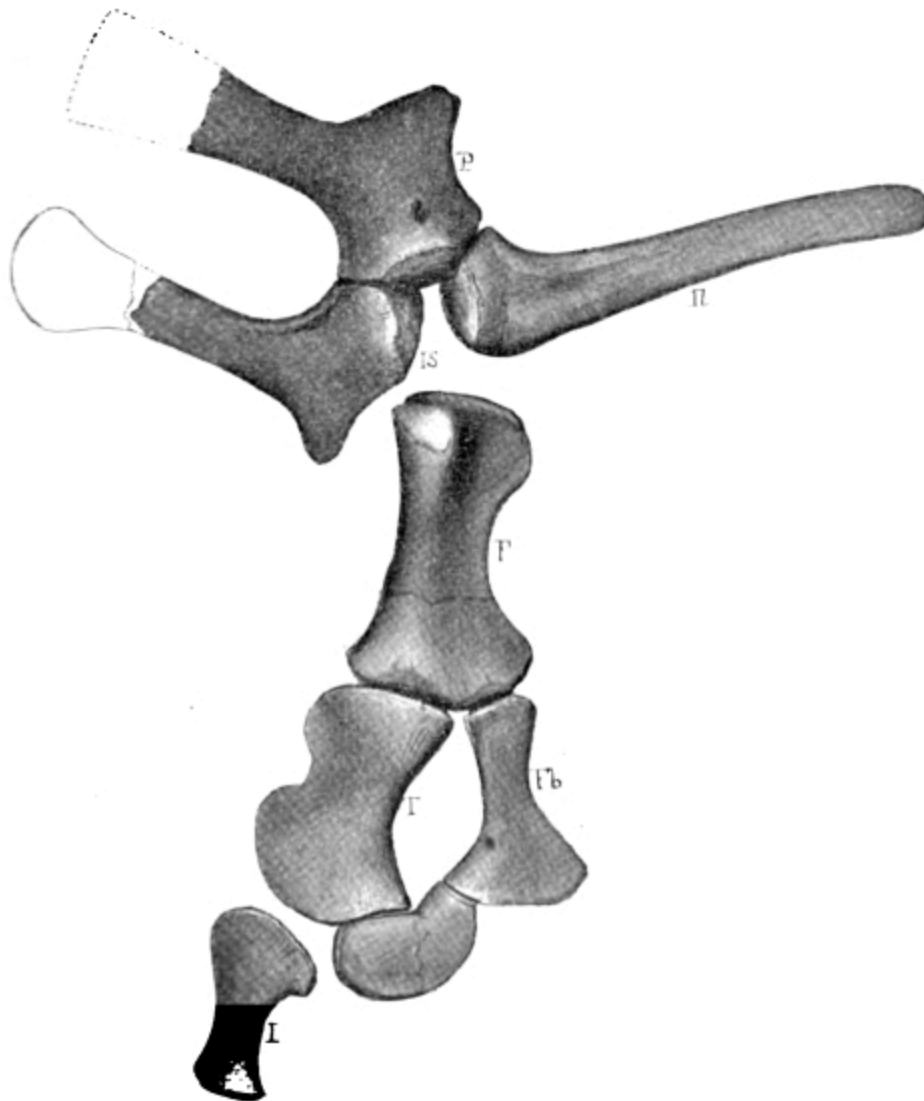


S. W. Williston, ad nat. del.

PLATE III.

Left hind paddle of *Clidastes velox* Marsh, two-thirds natural size. *Il*, ilium; *P*, pubis; *Is*, ischium; *F*, femur; *T*, tibia; *Fb*, fibula; *I*, first metatarsal.

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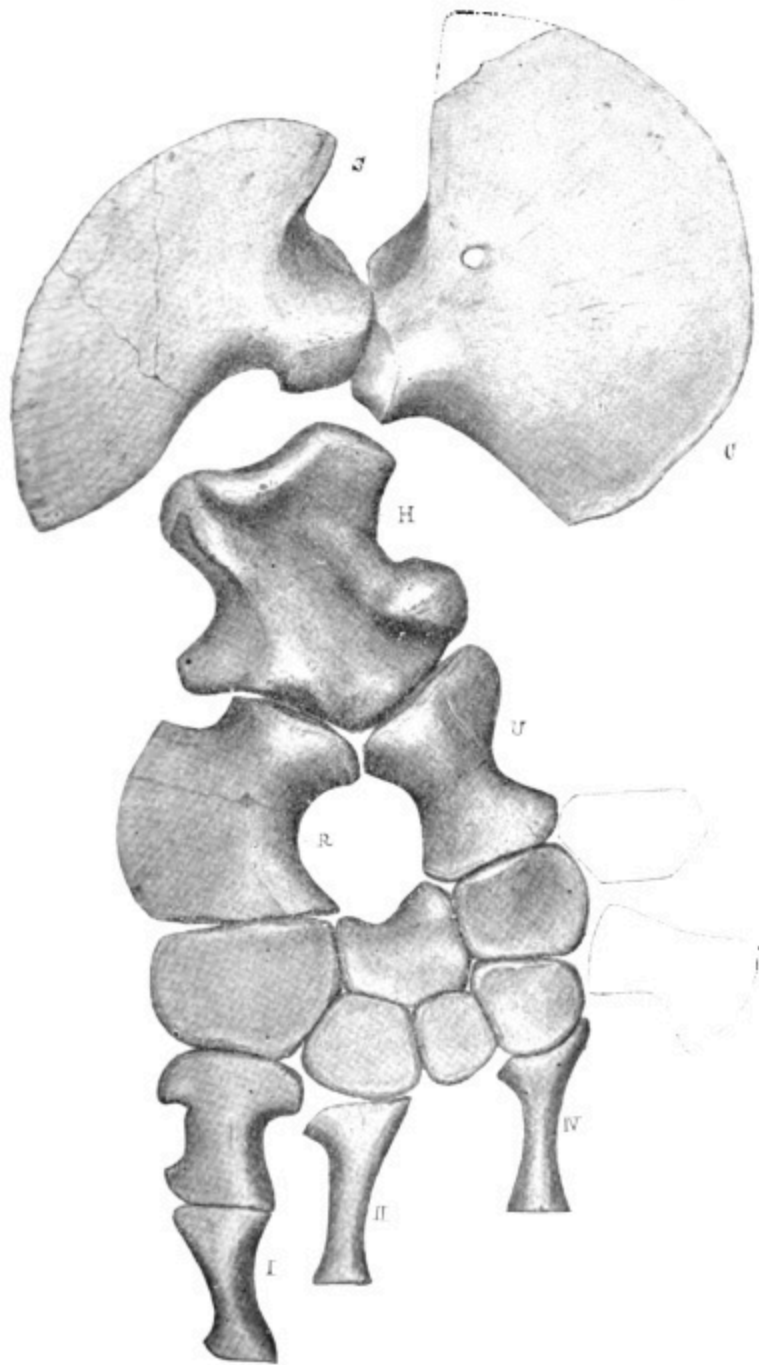


S. W. Williston, ad nat. del.

PLATE IV.

Right front paddle of *Clidastes Westii* Williston, one-third natural size.
S, scapula; C, coracoid; H, humerus; R, radius; U, ulna; I, IV, first, fourth
digits.

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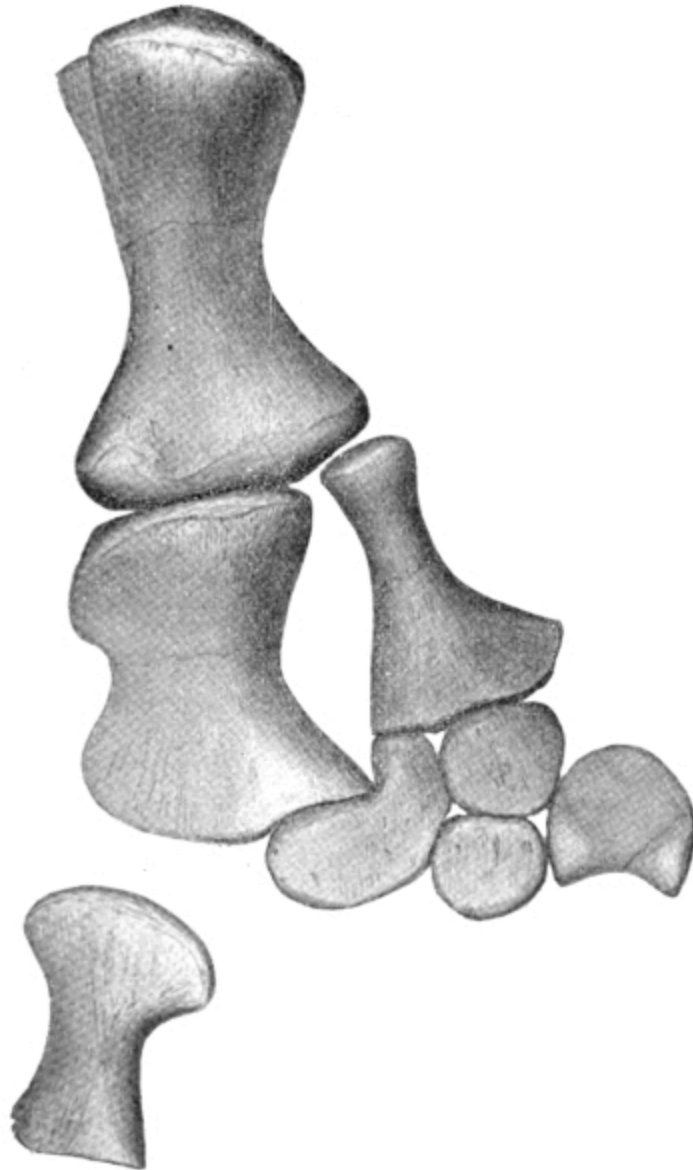
S. W. Williston, ad nat. del.



PLATE V.

Right hind paddle of *Clidastes Westii* Williston, one-half natural size.

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S. W. Williston, ad nat. del.



PLATE VI.

Eighteenth dorsal vertebra of *Clidastes Westii* Williston, natural size.
Fig. 1, centrum from behind; fig. 2, from below.

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Fig. 1.

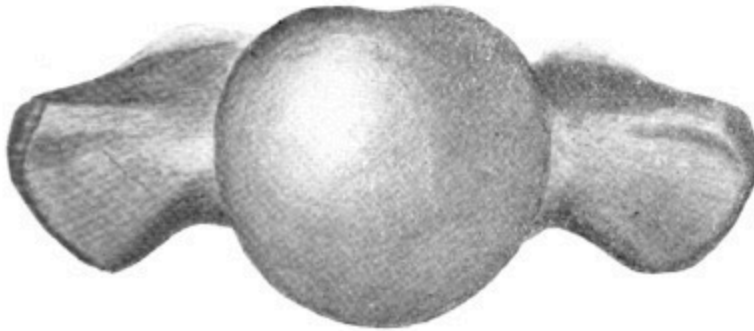


Fig. 2.

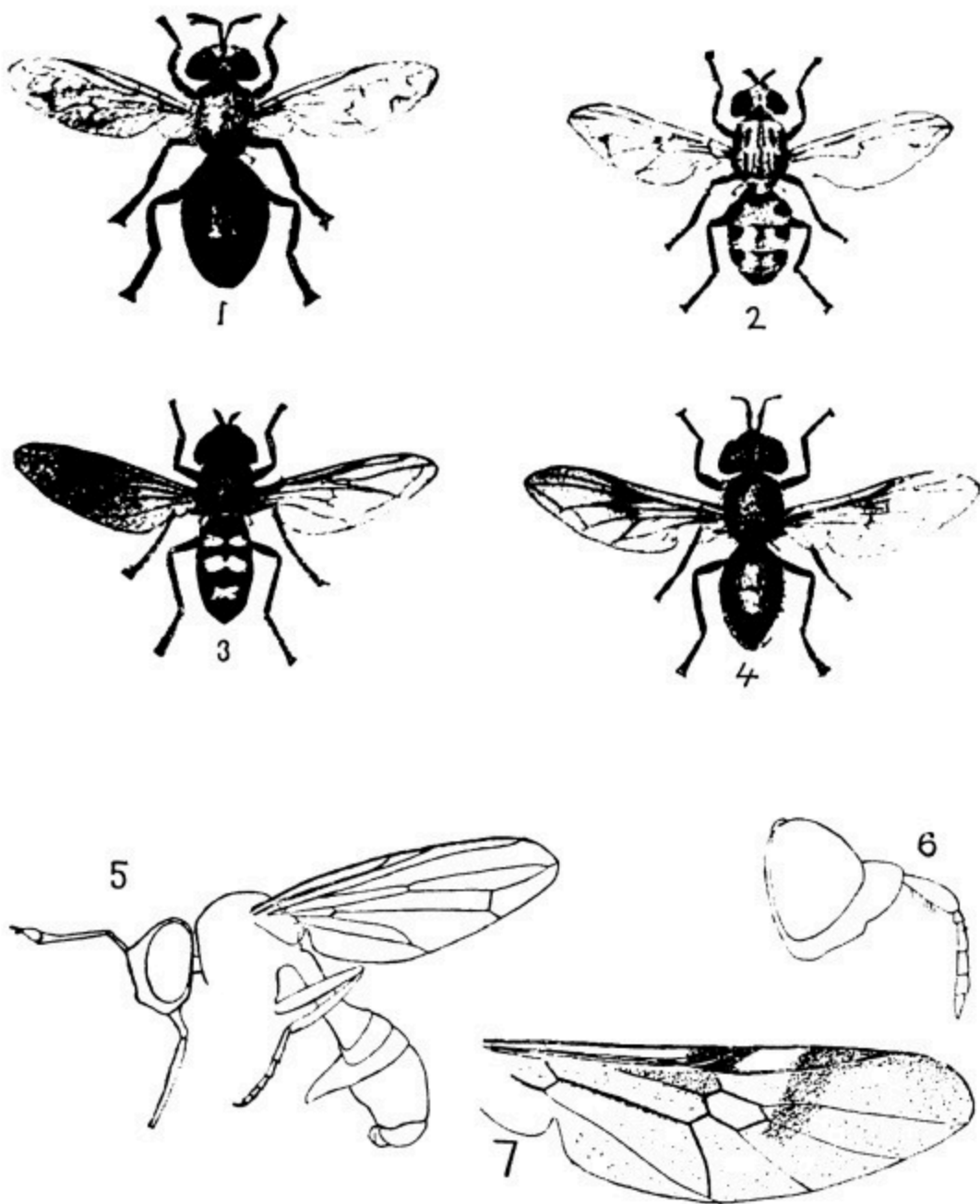


S. W. Williston, ad nat. del.



PLATE VII.

- Fig. 1, *Microdon megalogaster* Snow.
Fig. 2, *Brachyopa cynops* Snow.
Fig. 3, *Syrphus ruficauda* Snow.
Fig. 4, *Callicera montensis* Snow.
Fig. 5, *Tropidomyia bimaculata* Williston.
Fig. 6, *Promerisana nasuta* Macq.
Fig. 7, *Ancanthina hieroglyphica* Wiedemann.

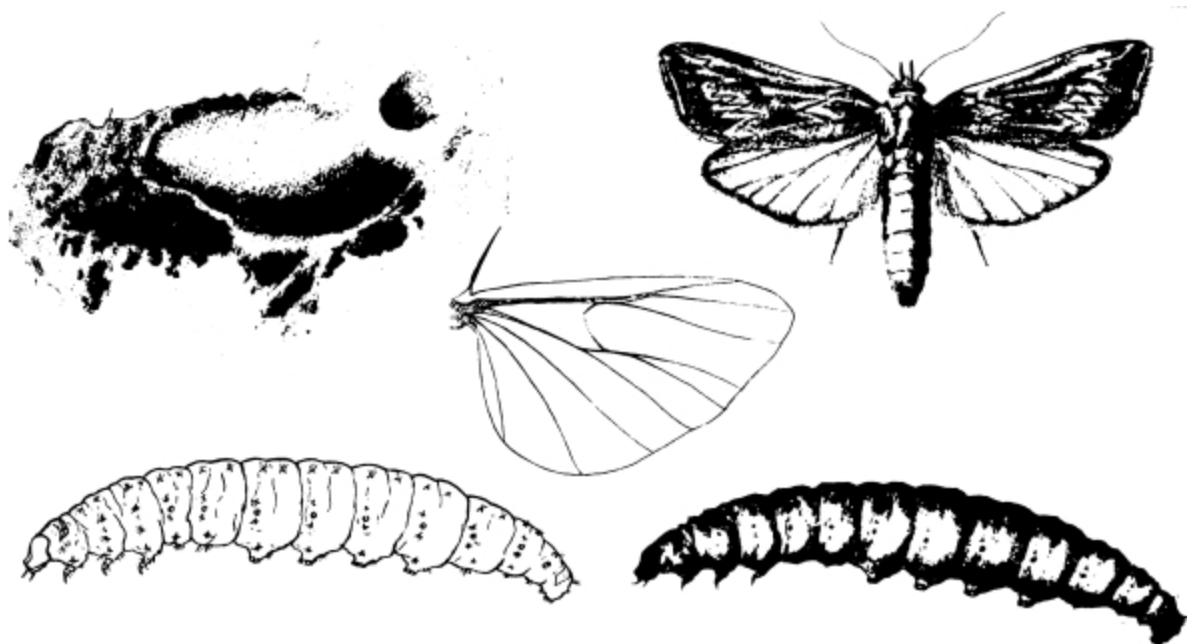


Mary Wellman and S. W. Williston, ad nat. del.

PLATE VIII.

Melitera dentata Grote; adult, silken cocoon and outer layer of dirt-masses held together by silken threads; larva (shaded); larva in outline showing position and number of tubercled hairs; hind wing of adult showing venation.

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Footnotes:

[1] This preoccupation rests, so far as I am aware, upon Marsh's statement. I can find no evidence of the name having been previously used.

[2] Amer. Naturalist, Dec. 1891, p. 1124. In this article the description of the foot-phalanges should read: "All are slender, except the second one in the third toe, and the second and third in the fourth toe, where they are scarcely longer than wide."

[3] The specimens in which I have seen the vertebral articulation show no co-ossification of the vertebrae: the facet for articulation being placed above the spines, and apparently formed by ossified ligaments.

[4] Proc. Bost. Soc. Nat. Hist., p. 253.

[5] Science, xvi, p. 262, Nov. 7, 1890.

[6] Amer. Journ. Sci. iii, June 1872.

[7] Amer. Journ. Sci. iii, pl. xi, f. 1, June, 1872.

[8] Amer. Journ. Sci. xix, pl. i, fig. 1, Jan., 1880.

[9] Leidy, Rep. U. S. Geol. Surv. Hayden, vol. i, p. 284.

[10] Bull. Su. Mus. Roy. S. Hist. Nat. d. Belg. i, p. 8, 1882.

[11] See Trans. Amer. Entom. Soc. xv, p. 243, for Part I.

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